

Canonical Field Structure of a Split Metric Bundle

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Abstract

The curvature of a single metric connection on $E = TM \oplus N$, read in its rank-stratified channels, carries a complete canonical field package for the vacuum Einstein constraint sector and an internal Yang–Mills-type gauge sector.

For a Riemannian manifold (M, h) and a metric vector bundle $N \rightarrow M$, set $E = TM \oplus N$. Fix the tangent block to be the Levi–Civita connection of h , and allow a normal metric connection together with a symmetric mixed tensor to vary. These data determine a metric connection $\nabla^\oplus$ on E , whose curvature $\mathcal{F}^\oplus$ splits into Gauss, mixed, and normal channels.

For rank $N = 1$, the normal channel is absent. Among natural linear metric vector-scalar contractions of the residual Gauss-mixed pair, the unique nondegenerate contraction is, after unit-normal trivialization, the fixed-metric vacuum Einstein momentum-Hamiltonian constraint pair. In the physical Cauchy regime, its vanishing gives smooth vacuum initial data and hence a Choquet-Bruhat–Geroch maximal globally hyperbolic vacuum spacetime development.

For rank $N \geq 2$, the normal curvature channel remains. In the first-jet admissible regime, non-Euler-neutral dependence is forced to factor through the assembled curvature. The trace classification then leaves the trace quadratic curvature density as the unique non-Euler-neutral representative; its projected Euler equation is Yang–Mills type. A fixed-product Legendre split gives canonical variables, a Hamiltonian, an internal Gauss constraint, and a normal-gauge momentum map.

The two normal-rank regimes thus identify the canonical field content carried by $\mathcal{F}^\oplus$, together with the phase-space data required before quantization questions need be posed.

Introduction

Let (M, h) be a Riemannian manifold, and let $N \rightarrow M$ be a rank- k vector bundle with fiber metric g_N . Set $E = TM \oplus N$ with product metric $h \oplus g_N$. Fix the tangent block to be the Levi–Civita connection of h , and allow the normal connection together with a symmetric mixed tensor to vary. These data determine a metric connection $\nabla^\oplus$ on E , with curvature $\mathcal{F}^\oplus$. Classical tangent–normal geometry is organized by the orthogonal decomposition $f^* \widetilde{TM} = f_* TM \oplus N_f M$, which yields the second fundamental form, the normal connection, and the Gauss–Codazzi–Ricci compatibility equations [4]. Chen–Li treat the same compatibility structure through the Cartan structural system and orthogonal splitting in the semi-Riemannian setting [3]. The curvature $\mathcal{F}^\oplus$ is the orthogonal compatibility curvature of the split data.

The rank of N determines which canonical field content remains visible in $\mathcal{F}^\oplus$. In rank one, the normal curvature channel vanishes, and the residual Gauss-mixed pair has a unique nondegenerate natural linear metric vector-scalar readout. This readout is identified below with the fixed-metric vacuum Einstein momentum-Hamiltonian constraint pair. In rank at least two, the normal curvature channel survives. First-jet admissibility forces non-Euler-neutral dependence through $\mathcal{F}^\oplus$, and trace classification leaves the trace quadratic curvature density; its projected Euler equation is Yang–Mills type.

Across these two regimes, the same structural discipline recurs. Curvature is paired by the defining metric representation of E ; scalar densities retain only non-Euler-neutral dependence carried by the assembled curvature; highest-order propagation is fixed by the split metric geometry rather than by field amplitude. Thus the split metric bundle supplies one metric connection, and the rank-stratified curvature channels determine the two canonical outputs.

In the rank-one case, $\mathfrak{so}(N) = 0$, so the normal curvature channel is absent. What remains of $\mathcal{F}^\oplus$ is the Gauss-mixed pair. The admissible linear metric contractions of this pair are classified, and the nondegenerate vector-scalar readout is unique up to componentwise scale. After a unit-normal trivialization, this readout is uniquely the fixed-metric vacuum Einstein momentum and Hamiltonian constraint pair. In the physical rank-one Cauchy regime, its vanishing makes (Σ, h, K) a smooth vacuum initial data set; the Choquet-Bruhat–Geroch theorem then gives the associated maximal globally hyperbolic vacuum spacetime development.

For $k \geq 2$, the normal curvature channel is nontrivial. The admissible first-jet regime forces non-Euler-neutral dependence to factor through the assembled curvature $\mathcal{F}^\oplus$. The trace classification then leaves the trace quadratic curvature density as the unique non-Euler-neutral representative, and the Euler equation on the admissible configuration space is the projected Yang–Mills-type equation $\Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0$.

A formal fixed-product split $M \simeq \mathbb{R} \times \Sigma$ gives the higher-rank equation a canonical interface. The Legendre transform produces variables

$$(A, K, p_A, p_K),$$

canonical symplectic form Ω_{can} , formal Riemannian Hamiltonian H_R , multiplier Hamiltonian H_{ext} , and internal normal-gauge momentum map J . The temporal normal connection acts as multiplier, and its equation is the internal Gauss constraint.

Thus the same orthogonal compatibility curvature produces two outputs in two normal-rank regimes: codimension-one vacuum spacetime initial data and higher-rank Yang–Mills-type gauge dynamics. All constructions are classical.¹

Sections 1 and 2 construct the split connection, its curvature channels, and the invariant configuration geometry. Sections 3–7 prove the higher-rank classification: first-jet complement exhaustion gives curvature dependence, derivative-curvature channels are excluded by order, nonaffine curvature functions are excluded by field-dependent principal symbols, the projected Euler operator and Noether identity are identified, and the trace quadratic density is shown to be the unique non-Euler-neutral representative in the stated regime. Section 8 develops the fixed-product canonical interface for this higher-rank field law. Section 9 treats the rank-one collapse, classifies the residual vector-scalar readouts, identifies the unique nondegenerate readout with the vacuum Einstein constraint pair, and hands the resulting data to the Choquet-Bruhat–Geroch Cauchy theorem. Appendices contain the covariant variation, projected Euler equation, symbol calculations, Noether identity, and contraction audit.

Conventions. For p -forms $\alpha, \beta \in \Omega^p(M, \mathfrak{so}(E))$, $\langle \alpha, \beta \rangle$ denotes the pointwise pairing induced by h and the fixed trace pairing on $\mathfrak{so}(E)$. Corresponding top-degree forms are written as

$$\text{tr}(\alpha \wedge \star \beta) = \langle \alpha, \beta \rangle dV_h.$$

¹No Hilbert space, quantum representation of the canonical algebra, quantum Gauss generator, quantum Hamiltonian, unitary quantum dynamics, Ward identity, semiclassical return theorem, or quantization of gravity is constructed. Canonical interface supplies only the classical data required before such later questions can be posed.

This convention is used for associated bundles after applying the relevant fiber metric. A symbol such as $\mathcal{L}_{\text{YM}}$ denotes the top-degree density/form being integrated. Boundary currents are written as differential forms; for example the Yang–Mills potential current is $\text{tr}(\delta\omega^\oplus \wedge \star\mathcal{F}^\oplus)$.

With the adjoint convention fixed by the L^2 pairing and Hodge metric, the ambient Yang–Mills expression is written equivalently as

$$D^\oplus \star \mathcal{F}^\oplus = 0 \quad \text{and} \quad (D^\oplus)^\dagger \mathcal{F}^\oplus = 0.$$

On $\mathcal{A}_\oplus$, the Euler–Lagrange equation is the projected equation $\Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0$. Equivalently, the second ambient form is the formal-adjoint rewriting of the first under this fixed convention before applying the admissible projection. This should be distinguished from the Noether identity

$$\pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_{\text{YM}}) = 0,$$

where $\mathcal{E}_{\text{YM}} = \Pi_{\mathcal{A}_\oplus}(-\star D^\oplus(\star\mathcal{F}^\oplus))$ is the admissible Euler–Lagrange operator and $\pi_{\mathfrak{so}(N)}$ is the normal-block projection associated with the restricted internal gauge symmetry.

1 Configuration Geometry

This section fixes the split metric bundle, the admissible metric connections, and the curvature channels used throughout.

1.1 Tangent–Normal Data

Definition 1.1 (Tangent–Normal Data). *Let (M, h) be an oriented smooth n -dimensional Riemannian manifold, with Levi–Civita connection ∇^h . Let $N \rightarrow M$ be an oriented rank- k vector bundle with fiber metric g_N , and define*

$$E = TM \oplus N, \quad g_E = h \oplus g_N.$$

Bundle E carries the product orientation. Thus the $\text{SO}(N)$ and $\text{SO}(E)$ frame bundles used below are globally defined. If the orientation hypotheses are dropped, the corresponding metric-only statements are obtained by replacing special-orthogonal structure groups by orthogonal structure groups and by omitting the explicitly oriented $n = 4$ Chern–Weil refinement.

The tangent block is fixed to ∇^h . Admissible variables are a metric normal connection and a symmetric mixed tensor.

1.2 Compatible Connections

Definition 1.2 (Orthogonal Compatibility Connection). *Let $\nabla^\perp$ be a metric connection on N , and let*

$$K \in \Gamma(\text{Sym}^2 T^*M \otimes N).$$

For $X \in TM$, write $K_X(Y) = K(X, Y)$ and let $K_X^ : N \rightarrow TM$ be its adjoint, characterized by*

$$\langle K_X(Y), \nu \rangle_{g_N} = h(Y, K_X^*(\nu)).$$

Define the orthogonal compatibility connection on E by

$$\nabla^\oplus = \nabla^h \oplus \nabla^\perp + \Psi,$$

where

$$\Psi(X) := \begin{pmatrix} 0 & -K_X^* \\ K_X & 0 \end{pmatrix}$$

acts on $T_x M \oplus N_x$.

Proposition 1.3 (Canonical Assembly). *For every pair $(\nabla^\perp, K)$ as above, there is a unique metric connection on $E = TM \oplus N$ whose tangent block is ∇^h , whose normal block is $\nabla^\perp$, and whose $TM \rightarrow N$ mixed block is K_X .*

Proof. Define

$$\nabla_X^\oplus(Y \oplus \nu) = (\nabla_X^h Y - K_X^* \nu) \oplus (\nabla_X^\perp \nu + K_X Y).$$

Product connection $\nabla^h \oplus \nabla^\perp$ is metric, and $\Psi(X)$ is skew-adjoint with respect to $h \oplus g_N$ by the definition of K_X^* . Hence $\nabla^\oplus$ is metric. If another metric connection has the same diagonal blocks and the same $TM \rightarrow N$ block, metricity forces its $N \rightarrow TM$ block to be $-K_X^*$. Thus it equals $\nabla^\oplus$. $\square$

Definition 1.4 (Admissible Configuration Space). *Let $\mathcal{C} \rightarrow M$ denote the affine bundle whose sections are orthogonal compatibility connections, equivalently pairs*

$$(\nabla^\perp, K) \in \Gamma(\mathcal{A}(N, g_N)) \times \Gamma(\text{Sym}^2 T^* M \otimes N),$$

with frozen tangent block ∇^h . The configuration space is

$$\mathcal{A}_\oplus := \Gamma(\mathcal{C}).$$

Principal Bundle Notation. Equivalently, the construction may be expressed on the principal $\text{SO}(E)$ -bundle $P(E) \rightarrow M$ associated to (E, g_E) , with principal connection $\omega^\oplus$, curvature $\mathcal{F}^\oplus$, and covariant exterior derivative $D^\oplus$. Under the standard correspondence between principal connections and associated covariant derivatives, $\omega^\oplus$ represents the metric connection $\nabla^\oplus$ above. Write $\text{rank}(E) = m = n + k$ throughout.

1.3 Admissible Variables and Projection

It is useful to separate the unrestricted mixed one-form from its admissible symmetric part. Let

$$B \in \Omega^1(M, \text{Hom}(TM, N)) \cong \Gamma(T^* M \otimes T^* M \otimes N).$$

Orthogonal compatibility imposes the symmetry condition

$$\Pi_{\text{skew}}(B) = 0, \quad \text{equivalently } B_{ij} = B_{ji}.$$

We write $K := \Pi_{\text{sym}}(B) \in \Gamma(\text{Sym}^2 T^* M \otimes N)$ for the admissible representative, with adjoint blocks as in Definition 1.2.

Definition 1.5 (Mixed Derivatives on the Admissible Bundle). *Let*

$$D^{h,\perp} : \Gamma(T^* M \otimes T^* M \otimes N) \rightarrow \Gamma(T^* M \otimes T^* M \otimes T^* M \otimes N)$$

denote the full product covariant derivative induced by $(\nabla^h, \nabla^\perp)$. In local coordinates:

$$(D^{h,\perp} K)_{\ell ij}^\alpha = \partial_\ell K_{ij}^\alpha - \Gamma_{\ell i}^r K_{rj}^\alpha - \Gamma_{\ell j}^r K_{ir}^\alpha + (A_\ell)^\alpha{}_\beta K_{ij}^\beta.$$

Under admissibility $K \in \Gamma(\text{Sym}^2 T^*M \otimes N)$, this operator is restricted to the symmetric mixed bundle and the notation $D^{h,\perp}$ is used for the full first derivative. The exterior Codazzi derivative is the antisymmetrization

$$(d^{h,\perp} K)_{ij\ell}^\alpha := (D^{h,\perp} K)_{ij\ell}^\alpha - (D^{h,\perp} K)_{jil}^\alpha.$$

Definition 1.6 (Admissible Variation Bundle and Euler Projection). *At an admissible connection $\omega^\oplus \in \mathcal{A}_\oplus$, the admissible variation bundle is*

$$T_{\omega^\oplus} \mathcal{A}_\oplus \cong \Omega^1(M, \mathfrak{so}(N)) \oplus \Gamma(\text{Sym}^2 T^*M \otimes N),$$

embedded in $\Omega^1(M, \mathfrak{so}(E))$ by

$$(\alpha, \kappa) \mapsto \begin{pmatrix} 0 & -\kappa^* \\ \kappa & \alpha \end{pmatrix}.$$

Let

$$\Pi_{\mathcal{A}_\oplus} : \Omega^1(M, \mathfrak{so}(E)) \longrightarrow T_{\omega^\oplus} \mathcal{A}_\oplus$$

denote the L^2 -orthogonal projection onto this embedded subbundle. Equivalently, for a block one-form

$$Q = \begin{pmatrix} Q_T & -Q_M^* \\ Q_M & Q_N \end{pmatrix},$$

the admissible projection keeps Q_N , keeps the symmetric part of Q_M in the two tangent slots, and discards the frozen TT block and the skew mixed part:

$$\Pi_{\mathcal{A}_\oplus} Q = (Q_N, \Pi_{\text{sym}} Q_M).$$

Remark 1.7 (Concrete Fiber Model of the Admissible Configuration Bundle). *Affine bundle $\mathcal{C}$ is realized inside the bundle of principal connections by the orthogonal compatibility ansatz. Setting $m = n + k$, its model fiber is affine over*

$$(\mathbb{R}^n)^* \otimes \mathfrak{so}(k) \oplus (\text{Sym}^2(\mathbb{R}^n)^* \otimes \mathbb{R}^k).$$

Thus the configuration space $\mathcal{A}_\oplus = \Gamma(\mathcal{C})$ is the full space of admissible compatible connections with frozen tangent block.

Dynamical Scope of the Admissible Class. No reduction to the $K \equiv 0$ subspace is made; the admissible configuration space retains both the normal connection and the symmetric mixed tensor.

1.4 Curvature Decomposition

Proposition 1.8 (Gauss–Codazzi–Ricci Curvature Blocks). *Let $F^\perp$ be the curvature of $\nabla^\perp$. For $\nabla^\oplus$, curvature decomposes into tangent, mixed, and normal blocks:*

$$\mathcal{F}^\oplus = \begin{pmatrix} \mathbf{G} & -\mathbf{C}^* \\ \mathbf{C} & \mathbf{S} \end{pmatrix}.$$

Here

$$\mathbf{G} = R^h - K^* \wedge K, \quad \mathbf{C} = d^{h,\perp} K, \quad \mathbf{S} = F^\perp - K \wedge K^*.$$

These are the Gauss, Codazzi, and Ricci curvature channels of the split metric connection.

Proof. Write $\nabla^\oplus = \nabla^{\text{prod}} + \Psi$, where $\nabla^{\text{prod}} = \nabla^h \oplus \nabla^\perp$. Then

$$\mathcal{F}^\oplus = F^{\nabla^{\text{prod}}} + D^{\nabla^{\text{prod}}} \Psi + \Psi \wedge \Psi, \quad F^{\nabla^{\text{prod}}} = \begin{pmatrix} R^h & 0 \\ 0 & F^\perp \end{pmatrix}.$$

Product covariant exterior derivative of the off-diagonal block is off-diagonal, with $TM \rightarrow N$ component $d^{h,\perp}K$ and adjoint component $-(d^{h,\perp}K)^*$. Since Ψ is off-diagonal,

$$\Psi \wedge \Psi = \begin{pmatrix} -K^* \wedge K & 0 \\ 0 & -K \wedge K^* \end{pmatrix}.$$

Adding the three block matrices gives the displayed identity. $\square$

Corollary 1.9 (Flatness and the GCR System). *Equation $\mathcal{F}^\oplus = 0$ is equivalent to the simultaneous vanishing of G , C , and S . In this form it is the Gauss–Codazzi–Ricci compatibility system for the split data.*

Proof. Directness of the block decomposition gives $\mathcal{F}^\oplus = 0$ if and only if each block vanishes. Resulting equations are the classical Gauss, Codazzi, and Ricci equations for a tangent-normal splitting; see Kobayashi–Nomizu [1, Vol. II] and the intrinsic GCR formulation of Chen–Li [2]. $\square$

Proposition 1.10 (Block Energy Splitting). *With respect to the trace pairing on $\mathfrak{so}(E)$ and the metric contraction on forms,*

$$\|\mathcal{F}^\oplus\|^2 = \|G\|^2 + 2\|C\|^2 + \|S\|^2.$$

Proof. Splitting

$$\mathfrak{so}(E) = \mathfrak{so}(TM) \oplus (\text{Hom}(TM, N) \oplus \text{Hom}(N, TM)) \oplus \mathfrak{so}(N)$$

is orthogonal under the trace pairing. Hence the diagonal and off-diagonal blocks have no cross terms. Mixed channel occurs in the two adjoint off-diagonal slots, giving the coefficient 2. $\square$

1.5 Holonomy Obstruction and Curvature Sourcing

For a connection, curvature is the infinitesimal obstruction to flat parallel transport. For the metric connection $\nabla^\oplus$, this obstruction is the assembled two-form

$$\mathcal{F}^\oplus \in \Omega^2(M, \mathfrak{so}(E)).$$

Its Gauss, mixed, and normal blocks are the components of one $\mathfrak{so}(E)$ -valued curvature, not separate sources with freely chosen weights.

A scalar density has *holonomy-obstruction dependence* if its non-Euler-neutral dependence on the admissible variables $(\nabla^\perp, K)$ factors through $\mathcal{F}^\oplus$, the fixed metric data, and the trace pairing induced by the defining orthogonal representation of E . In this sense curvature-sourcing is not an additional preference for a Yang–Mills form. It is the requirement that the field law read the tangent-normal connection through its infinitesimal holonomy obstruction.

The first-jet theorem below proves this curvature-sourcing conclusion for non-Euler-neutral densities satisfying the stated semilinear structural regime. Terms that are local or gauge-natural under weaker hypotheses, such as independent K -potentials, first-jet complement densities, raw connection-coefficient expressions, or independently weighted block laws, are then excluded either by complement exhaustion, by differential order, by semilinearity, or by the trace classification.

2 Gauge-Natural Configuration Geometry

The higher-rank classification uses the pairing on $\mathfrak{so}(E)$ -valued curvature induced by the defining metric representation of E . Independently weighted pairings on the split curvature blocks define a different admissibility class. The trace-induced pairing fixed here also determines the kinetic metric used by the fixed-product Legendre split.

2.1 Gauge-Natural Structure

Internal normal gauge transformations preserve the frozen tangent block, the admissible mixed symmetry, and the curvature-channel decomposition. Scalar densities used below are local gauge-natural functions of the admissible configuration jets.

Let $SO(M, h)$ and $SO(N, g_N)$ denote the oriented orthonormal frame bundles fixed in Definition 1.1. Unless an oriented refinement is explicitly named, the invariant-theory statements below are made in the metric-only category, with algebraic tests against the full $O(n) \times O(k)$ action on model fibers. Thus orientation tensors are not admissible inputs except in statements that explicitly invoke the oriented $SO(n)$ refinement.

Vertical principal bundle automorphisms over id_M form the internal gauge group

$$\mathcal{G}_N := \Gamma(\text{SO}(N)).$$

Proposition 2.1 (Internal Gauge Equivariance). *Admissible configuration space $\mathcal{A}_\oplus$ is invariant under the restricted gauge group $\mathcal{G}_N$. Under the inclusion $u \mapsto \text{diag}(I_{TM}, u) \in \Gamma(\text{SO}(E))$, the principal action*

$$\omega^\oplus \mapsto u^{-1}\omega^\oplus u + u^{-1}du$$

preserves the frozen tangent block and sends $\mathcal{F}^\oplus$ to $u^{-1}\mathcal{F}^\oplus u$. Consequently the block decomposition into $\mathfrak{G}$, $\mathfrak{C}$, and $\mathfrak{S}$ is preserved by internal gauge.

Proof. Embedded gauge transformations act as the identity on TM and orthogonally on N . Therefore they fix the tangent Levi-Civita block, transform $\nabla^\perp$ by the usual connection law, and transform the mixed block tensorially by $K \mapsto u^{-1}K$. Tensorial transformation preserves the symmetry condition $\Pi_{\text{skew}}(B) = 0$. Curvature covariance for principal connections gives $\mathcal{F}^\oplus \mapsto u^{-1}\mathcal{F}^\oplus u$, and the splitting $E = TM \oplus N$ is preserved by the block-diagonal action. Hence the three curvature channels transform within their own block types. $\square$

Lemma 2.2 (Sobolev Smoothness of the Restricted Gauge Action). *Let $s > n/2 + 1$. Then the map*

$$H^{s+1}(M, \text{SO}(N)) \times \mathcal{A}_\oplus^s \rightarrow \mathcal{A}_\oplus^s, \quad (u, \omega^\oplus) \mapsto u^{-1}\omega^\oplus u + u^{-1}du$$

with block embedding $\text{diag}(I_{TM}, u)$ is smooth.

Proof. For $s > n/2 + 1$, $H^{s+1}(M, \text{SO}(N))$ is a Hilbert Lie group; inversion and multiplication are smooth in this model. Sobolev multiplication is continuous for $s > n/2$, and $d : H^{s+1} \rightarrow H^s$ is continuous. Therefore $(u, \omega) \mapsto u^{-1}\omega u + u^{-1}du$ is smooth as a map into H^s -connections. Since the block embedding $\text{SO}(N) \hookrightarrow \text{SO}(E)$, $u \mapsto \text{diag}(I_{TM}, u)$, is a smooth bundle morphism preserving the frozen tangent block, the action restricts smoothly to $\mathcal{A}_\oplus^s$. This is the standard Sobolev gauge-group smoothness mechanism in Yang-Mills settings; see [22, 25]. $\square$

Definition 2.3 (Gauge-Natural Local Scalar Density). *Let $\mathcal{G}_{\text{loc}}$ be the local orthogonal-gauge groupoid whose morphisms are pairs*

$$\Phi = (\varphi, u),$$

where φ is an orthonormal frame morphism on TM and u is a local $SO(N)$ gauge morphism on N . An order- r gauge-natural scalar density is a natural transformation

$$\mathcal{L} : J^r \mathcal{C} \longrightarrow \Lambda^n T^* M$$

commuting with the $\mathcal{G}_{\text{loc}}$ -action on jets and the induced action on densities, in the sense of the gauge-natural framework of Kolář–Michor–Slovák [8, Ch. XII]. On local algebraic jets,

$$(A, B, \partial A, \partial B) \mapsto (A^\Phi, B^\Phi, \partial A^\Phi, \partial B^\Phi),$$

and admissibility of the mixed subbundle is preserved:

$$\Pi_{\text{skew}}(B) = 0 \iff \Pi_{\text{skew}}(B^\Phi) = 0.$$

The groupoid $\mathcal{G}_{\text{loc}}$ tests naturality of local scalar densities; the dynamical gauge group acting on configurations remains the internal normal gauge group $\mathcal{G}_N$.

Remark 2.4 (Analytic and Boundary Convention). When functional-analytic statements are made, $s > n/2+1$, $\mathcal{A}_\oplus^s$ denotes the Hilbert completion of $\mathcal{A}_\oplus$, and the restricted gauge group is $H^{s+1}(M, SO(N))$. Integration by parts is always taken either on compact manifolds without boundary or for compactly supported interior variations. Unless a global Fredholm statement is explicitly named, the Noether and curvature-momentum identities are local.

2.2 Affine Model

Proposition 2.5. $\mathcal{A}_\oplus$ is an affine space modeled on the vector subspace $V \subset \Omega^1(M, \mathfrak{so}(E))$, where vectors in V vanish on the $TM \rightarrow TM$ tangent block.

Proof. Let $\nabla_0^\oplus \in \mathcal{A}_\oplus$. For any $\nabla^\oplus \in \mathcal{A}_\oplus$, set $a := \nabla^\oplus - \nabla_0^\oplus \in \Omega^1(M, \mathfrak{so}(E))$. Admissibility fixes the $TM \rightarrow TM$ block (frozen Levi–Civita block), while the normal variation $\delta\nabla^\perp$ and mixed variation δB remain. Imposing $\Pi_{\text{skew}}(\delta B) = 0$ gives the admissible mixed tangent space $V_{\text{mix}} = \Gamma(\text{Sym}^2 T^* M \otimes N)$. Hence a ranges over a linear subspace $V \subset \Omega^1(M, \mathfrak{so}(E))$, and $\mathcal{A}_\oplus = \nabla_0^\oplus + V$ is affine. $\square$

2.3 Canonical Trace Pairing

For dynamic offsets $\dot{\nabla}_1, \dot{\nabla}_2 \in T_{\nabla^\oplus} \mathcal{A}_\oplus \cong V$, the canonical inner product is defined via global integration:

$$g_{L^2}(\dot{\nabla}_1, \dot{\nabla}_2) := \int_M \langle \dot{\nabla}_1, \dot{\nabla}_2 \rangle dV_h.$$

Lemma 2.6 (Invariant Bilinear Forms). Let $m = n + k \geq 3$. For $m \neq 4$, the space of Ad -invariant symmetric bilinear forms $\text{Sym}^2(\mathfrak{so}(m)^*)^{\text{Ad}}$ is one-dimensional, spanned uniquely by the trace form $B(X, Y) = \text{tr}(XY)$. For $m = 4$, the algebra decomposes as $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, and the invariant space is two-dimensional, spanned by the independent Killing forms of the two ideals.

Proof. For a simple Lie algebra $\mathfrak{g}$, the space of Ad -invariant symmetric bilinear forms is one-dimensional. Since $\mathfrak{so}(m)$ is compact simple for $m \geq 5$ and $m = 3$, any such form is a scalar multiple of the Cartan–Killing form $\kappa(X, Y) = (m - 2)\text{tr}(XY)$. Thus, $\dim(\text{Sym}^2(\mathfrak{so}(m)^*)^{\text{Ad}}) = 1$, uniquely spanned up to scale by the trace pairing $B(X, Y) = \text{tr}(XY)$.

For the exceptional case $m = 4$, the Lie algebra is semisimple $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$. Schur’s Lemma applies independently to the adjoint representation of each simple ideal. Therefore, any

invariant symmetric bilinear form is a linear combination $a\kappa_L + b\kappa_R$, where $\kappa_{L,R}$ are the Killing forms of the respective $\mathfrak{su}(2)$ ideals, making the invariant space 2-dimensional; see Hall (2015), Chapter 10 [16]. $\square$

Definition 2.7 (Representation-Induced Pairing). *An admissible invariant bilinear form on $\mathfrak{so}(E)$ is required to arise functorially from the defining metric representation $O(E) \subset \text{End}(E)$. Equivalently, it is natural with respect to all orthogonal bundle isomorphisms $E \rightarrow E'$, including orientation-reversing ones. This is a representation-induced $O(E)$ -naturality condition on the pairing, even though the active gauge group of the admissible configuration space is the internal $\text{SO}(N)$ group.*

Remark 2.8 (Pairing Source). *Before a field law is chosen, the defining metric representation of E is the available structure for pairing $\mathfrak{so}(E)$ -valued curvature data; this makes the trace class induced by that representation the natural admissible pairing.*

Lemma 2.9 (Representation-Induced Restriction Principle). *Let $V = \mathbb{R}^m$ with its Euclidean structure. Any symmetric bilinear form on $\mathfrak{so}(V)$ that is natural under orthogonal bundle isomorphisms and induced from the ambient endomorphism representation is the restriction of an $O(V)$ -invariant symmetric bilinear form on $\text{End}(V)$. Hence it is a scalar multiple of $\text{tr}(XY)$ on $\mathfrak{so}(V)$.*

Proof. Naturality under orthogonal bundle isomorphisms gives conjugation invariance under $O(V)$. By the representation-induced hypothesis, the form is obtained by restricting an $O(V)$ -invariant symmetric bilinear form on $\text{End}(V)$. Such forms are generated by $\text{tr}(XY)$ and $\text{tr}(X)\text{tr}(Y)$. Restriction to $\mathfrak{so}(V)$ removes the trace-trace channel since $\text{tr}(X) = 0$, leaving only a scalar multiple of $\text{tr}(XY)$. $\square$

Theorem 2.10 (Structural Invariant Pairing Classification). *Let $E \rightarrow M$ be an orthogonal rank- m bundle with $m \geq 3$. In the representation-induced admissible class of Definition 2.7, the $O(E)$ -natural symmetric bilinear forms on $\mathfrak{so}(E)$ compatible with the defining orthogonal representation form a one-dimensional trace class:*

$$B_{\text{adm}}(A, B) = c \text{tr}(A \circ B), \quad c \neq 0.$$

For $m = 4$, the algebraic $\text{SO}(4)$ -semisimple freedom $a\kappa_L + b\kappa_R$ is removed by the representation-induced $O(E)$ -naturality condition, so only the trace-restriction class is admissible.

Proof. For $m \neq 4$, the Lie algebra $\mathfrak{so}(m)$ is simple and invariant symmetric bilinear forms are one-dimensional. For $m = 4$, write $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$, so $\text{SO}(4)$ -invariant forms are $a\kappa_L + b\kappa_R$. Orientation-reversing orthogonal transformations interchange the two $\mathfrak{su}(2)$ ideals. By Lemma 2.9, only the representation-induced orthogonal trace class is functorial under all orthogonal bundle isometries in families. Therefore the trace-induced class is unique up to scale in all dimensions. $\square$

Proposition 2.11 (Trace-Class Pairing Uniqueness Up to Scale). *Under the representation-induced requirement, the admissible invariant pairing on $\mathfrak{so}(E)$ is unique up to scale:*

$$B_{\text{adm}}(A, B) = c \text{tr}(A \circ B), \quad c \neq 0.$$

Proof. This is immediate from Lemma 2.9 and Theorem 2.10: the representation-induced admissible class is one-dimensional and generated by the orthogonal trace class. $\square$

Remark 2.12 (Pairing Normalization Convention). *In Proposition 2.11, the multiplicative constant is a fixed normalization convention (equivalently one may set $c = -1$ and use $-\text{tr}$). This convention is not part of the uniqueness proof. Its scale is global for the classified covariant density together with the induced configuration-space metric and is absorbed into the overall normalization parameter, not into the Yang–Mills coupling g alone.*

Theorem 2.13 (Trace-Induced Kinetic Metric Uniqueness). *After imposing the ambient trace-induced pairing requirement with positive ultralocal pairing on admissible velocity fibers, on the admissible configuration space $\mathcal{A}_\oplus^s$, the induced ultralocal invariant kinetic metric obtained by restricting the trace pairing class on $\Omega^1(M, \mathfrak{so}(E))$ to the admissible tangent bundle is unique up to a global scale. After fixing the normalization convention of Remark 2.12, this yields the canonical L^2 metric defining $T(\dot{\omega}^\oplus) = \frac{1}{2}\|\dot{\omega}^\oplus\|_{L^2}^2$.*

Proof. At each $x \in M$, admissible tangent data decomposes as

$$V_x \cong (T_x^*M \otimes \mathfrak{so}(k)) \oplus (\text{Sym}^2 T_x^*M \otimes N_x).$$

Here the first summand is the normal sector and the second is the admissible mixed sector. Under $O(n) \times O(k)$ these are non-isomorphic finite-dimensional representations, so any invariant bilinear cross-term between them vanishes by complete reducibility (compact isotropy) and Schur-type orthogonality.

Hence an ultralocal invariant fiber metric is determined by invariant pairings on each summand. In general this gives one scale per irreducible channel. Single-scale uniqueness here is the admissibility condition that the kinetic form is the restriction of the ambient trace-induced Yang–Mills pairing on

$$\Omega^1(M, \mathfrak{so}(E))$$

to the admissible subbundle. Under this restriction, both sectors are measured by the same representation-induced trace class from Definition 2.7; by Lemma 2.6 and Proposition 2.11, this pairing is unique up to one global constant. Integrating over M yields exactly a global-scale multiple of the canonical L^2 metric, with normalization fixed only by convention. $\square$

Define the geometric kinetic quantity

$$T(\dot{\nabla}^\oplus) := \frac{1}{2}\|\dot{\nabla}^\oplus\|_{L^2}^2.$$

3 Admissible Curvature Densities

Higher-rank classification begins with local gauge-natural first-jet densities in the split variables $(\nabla^\perp, K)$. The first-jet regime below does not assume curvature-sourcing. Gauge removal and principal-symbol rigidity eliminate the non-curvature complements, leaving a representative that factors through the assembled curvature $\mathcal{F}^\oplus$. The trace classification then leaves the trace quadratic curvature density as the unique non-Euler-neutral representative in the classified regime.

Three nested restrictions are used below. Baseline admissibility fixes the structural PDE class. Universal first-jet admissibility derives curvature dependence within that class. The representation-induced curvature-density regime then classifies the remaining curvature representatives. The terminology separates genuine competitors from terms invisible to the Euler operator, removable by equivalence, or outside the fixed principal-symbol class.

3.1 Higher-Rank Regime

Remark 3.1 (Frozen TT -Block Scope). *Fixing the Levi–Civita tangent block is part of the configuration object. Unfreezing the TT block introduces tangent-sector and metric-variation degrees of freedom and changes the classification problem; that enlarged regime is outside the present analysis.*

Remark 3.2 (Standing Nontrivial Gauge Regime). *The nontrivial orthogonal-compatibility gauge regime is*

$$n \geq 2, \quad k \geq 2.$$

Here $\Lambda^2 T^*M$ is nontrivial and the normal gauge algebra $\mathfrak{so}(k)$ is nontrivial. For $n = 1$, the curvature two-form vanishes identically. For $k = 1$, the normal gauge algebra is trivial and the codimension-one problem belongs to Section 9.

Definition 3.3 (Semilinear Principal-Symbol Structure). *Let $V \rightarrow M$ be the admissible variation bundle. A nonlinear Euler–Lagrange operator in this framework has semilinear principal-symbol structure if its principal symbol descends to a fixed $O(n) \times O(k)$ -equivariant bundle morphism*

$$\sigma : S^*M \times V \rightarrow V$$

*over S^*M , independent of configuration amplitude.*

Remark 3.4 (Structural Principal Type). *The second-order and semilinear principal-symbol conditions isolate field laws whose highest-order propagation structure is determined by the split metric geometry alone, independently of field amplitude. This is the PDE-side form of the same structural restriction that appears as curvature-sourcing on the Lagrangian side.*

Definition 3.5 (Baseline Admissibility Regime). *Baseline admissibility on $\mathcal{A}_\oplus$ specifies the local gauge-natural covariant scalar densities to which the exclusion arguments apply. A density belongs to this baseline regime if it satisfies:*

1. *locality of the variational density;*
2. *gauge-naturality (orthonormal-frame naturality and restricted $\mathcal{G}_N$ -equivariance);*
3. *Euler–Lagrange order at most two in the covariant field variables $(\nabla^\perp, K)$, modulo Euler-neutral additions and degenerate leading dependence;*
4. *semilinear principal-symbol structure in the sense of Definition 3.3.*

Second-order control is part of the structural PDE regime; it is not a consequence of locality or gauge-naturality alone. This baseline regime is not, by itself, asserted to force a single scalar density. Higher-rank uniqueness below uses the first-jet regime of Definition 3.7 and the curvature-density classification of Definition 3.9.

Definition 3.6 (Euler-Neutral Additions). *A local addition to a Lagrangian is called Euler-neutral if it does not change the Euler–Lagrange operator in the boundary regime (closed manifold or compactly supported variations). Total divergences and the oriented quadratic Chern–Weil density $\text{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus)$ in dimension four are Euler-neutral. Higher Chern–Weil forms are outside the curvature-density regime classified here.*

Definition 3.7 (Universal First-Jet Admissibility Regime). *In the higher-rank regime $n \geq 2$, $k \geq 2$, a local scalar density on $\mathcal{A}_\oplus$ belongs to the universal first-jet admissibility regime if it satisfies:*

1. *it is a gauge-natural scalar density of first jet order in the variables $(\nabla^\perp, K)$;*
2. *it is natural under orthonormal tangent-frame changes and equivariant under the restricted internal normal gauge group $\mathcal{G}_N$;*
3. *its Euler–Lagrange operator has differential order at most two;*

4. its second-order principal symbol is field-independent semilinear in the sense of Definition 3.3;
5. each non-Euler-neutral leading component is principal-effective on admissible first-jet directions;
6. equivalence is taken modulo constant background densities, total divergences, Euler-neutral additions, degenerate leading dependence, and terms already represented by functions of $\mathcal{F}^\oplus$, which are passed to the curvature-density classification below.

This regime is a first-jet regime. It does not assume that the density factors through $\mathcal{F}^\oplus$.

Remark 3.8 (Curvature-Sourced Representation Principle). *Representation-induced structure is stronger than invariance under the active internal $\mathrm{SO}(N)$ gauge group. Once the universal first-jet theorem has reduced non-Euler-neutral dependence to the infinitesimal holonomy obstruction $\mathcal{F}^\oplus \in \Omega^2(M, \mathfrak{so}(E))$, the curvature-density classification reads the field law functorially through the defining orthogonal representation of E . Independently weighting the assembled tangent, mixed, and normal curvature blocks is not a law read from $\mathcal{F}^\oplus$ through that representation.*

Definition 3.9 (Representation-Induced Curvature-Density Regime). *In the higher-rank regime $n \geq 2$, $k \geq 2$, the representation-induced curvature-density regime is the curvature-representative class reached after first-jet complement exhaustion. A density belongs to this regime if it satisfies the baseline admissibility regime and, in addition:*

1. all non-Euler-neutral dependence on the dynamical variables factors through the full assembled curvature

$$\mathcal{F}^\oplus \in \Omega^2(M, \mathfrak{so}(E)),$$

and the fixed background metric data, not through independently weighted curvature blocks or independent zeroth-order potentials in A , K , or blockwise variables;

2. its Lie-algebra pairing is induced from the defining orthogonal representation of E , equivalently from the $O(E)$ -natural trace class of Definition 2.7;
3. its curvature dependence is a smooth representation-induced invariant function of $\mathcal{F}^\oplus$, modulo Euler-neutral additions in the sense of Definition 3.6;
4. the only oriented topological refinement admitted is the quadratic $n = 4$ Chern–Weil density $\mathrm{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus)$; higher Chern–Weil densities are outside this theorem’s density category;
5. its non-Euler-neutral curvature dependence is principal-effective on the admissible curvature-symbol image: dependence whose second curvature differential is invisible on all admissible principal curvature variations is outside this density category unless it is constant or Euler-neutral;
6. its second-order top-level Euler–Lagrange symbol satisfies the field-independent semilinearity condition of Definition 3.3.

This is the curvature-density category in which the trace pairing and curvature invariant are classified. Curvature factorization for first-jet densities is supplied separately by Theorem 3.39.

Finite-order naturality, orthogonal invariant contractions, and Chern–Weil alternatives are standard inputs. The first-jet regime controls non-curvature complements. The representation-induced curvature-density regime fixes the source object $\mathcal{F}^\oplus$, the trace pairing class, and the curvature-symbol category.

Classification Path. Locality gives finite-jet locality. Second-order control reduces nondegenerate leading channels to first jets. Theorem 3.39 removes the non-curvature first-jet complements in the stated regime. Derivative-curvature channels are excluded by order, and nonaffine curvature functions are excluded by field-dependent principal symbols. What remains, modulo Euler-neutral terms and the stated equivalence relation, is the trace quadratic curvature density. Kinetic pairing rigidity is separate and follows from the representation-induced pairing classification in Section 2.

3.2 Exclusion Mechanisms

After first-jet complement exhaustion, three curvature-density competitor classes remain. Independent algebraic potentials fail holonomy-obstruction dependence. Genuine derivative-curvature dependence leaves the second-order structural PDE class. Nonaffine functions of curvature invariants make the second-order principal coefficients depend on the background curvature.

Proposition 3.10 (Lower-Order Potentials and Curvature Sourcing). *In the representation-induced curvature-density regime, a nonconstant algebraic potential depending independently on A , K , or separated curvature blocks lies outside the admitted curvature-sourced class. If such a term is added to the curvature density, its first variation contributes a lower-order force not generated by the assembled curvature divergence and is not Euler-neutral unless the potential is constant on admissible fibers.*

Proof. An algebraic potential such as a nonconstant $O(n) \times O(k)$ -invariant function of K is local and gauge-natural, but it does not factor through the assembled curvature $\mathcal{F}^\oplus$. Its variation is a zeroth-order functional derivative in the admissible field variables. This term is not represented by the curvature divergence $\Pi_{\mathcal{A}^\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus)$, and it changes the Euler–Lagrange operator unless the potential is constant on admissible fibers. Hence it is a lower-order force added to the curvature law, not a representative of the curvature-sourced density class. $\square$

Theorem 3.11 (Finite-Jet Locality). *Under Definition 3.5, local gauge-natural scalar densities factor through finite jet bundles.*

Proof. This is the gauge-natural operator framework of Kolář–Michor–Slovák (Chapter XII), together with the Peetre–Slovák finite-order theorem in this setting [8]. Here the needed form is the finite-order locality theorem for gauge-natural operators (KMS Thm. 19.7 together with the natural-operator reduction in Chapter XII). $\square$

Lemma 3.12 (Differential-Order Counting). *For a local Lagrangian $L = L(j^r \omega^\oplus)$, the Euler–Lagrange operator has order at most $2r$. In particular:*

- if $L = P(\mathcal{F}^\oplus)$ with no derivatives of $\mathcal{F}^\oplus$, then $\mathcal{E}_L$ is second-order in $\omega^\oplus$;
- if L has nondegenerate leading dependence on $D^\oplus \mathcal{F}^\oplus$ or higher curvature jets, then $\mathcal{E}_L$ has order exceeding two in general and is excluded by the second-order requirement in Definition 3.5.

Proof. Standard variational order counting gives $\text{ord}(\mathcal{E}_L) \leq 2r$. Since $\mathcal{F}^\oplus$ is first-order in $\omega^\oplus$, curvature-only channels depend on first jets of $\omega^\oplus$, hence $r = 1$ and the Euler–Lagrange operator is second-order. Nondegenerate dependence on $D^\oplus \mathcal{F}^\oplus$ raises the jet order to $r = 2$, and higher curvature jets raise it further. Such leading channels therefore produce fourth-order or higher Euler operators in general and violate the second-order part of Definition 3.5. $\square$

Theorem 3.13 (First-Jet Top-Order Reduction). *Under the second-order structural PDE requirement in Definition 3.5, nondegenerate top-order channels in the orthogonal-compatibility model reduce, modulo Euler-neutral additions and degenerate leading dependence, to the first-jet class of $(\nabla^\perp, B)$.*

Proof. Using Theorem 3.11 together with the order-counting result (Lemma 3.12), a genuine top-order dependence beyond first jets produces Euler–Lagrange channels that leave the second-order structural PDE class. Dependence on second or higher jets introduces curvature-derivative channels such as $D^\oplus \mathcal{F}^\oplus$, whose Euler–Lagrange operators are generically fourth order by Lemma 3.12. Euler-neutral additions and terms whose apparent top dependence has zero Euler contribution are removed by the stated equivalence. Hence nondegenerate higher-jet leading dependence contradicts Definition 3.5. $\square$

Theorem 3.14 (General Derivative-Curvature Jet-Order Exclusion). *Let L be a local gauge-natural Lagrangian whose nondegenerate leading channel depends on curvature derivatives $(D^\oplus)^q \mathcal{F}^\oplus$ with $q \geq 1$, modulo Euler-neutral additions. Then the associated Euler–Lagrange operator has differential order strictly greater than two in $\omega^\oplus$. Therefore such genuine derivative-curvature leading channels are excluded by the second-order structural PDE requirement in Definition 3.5.*

Proof. Since $\mathcal{F}^\oplus$ is first-order in $\omega^\oplus$, $(D^\oplus)^q \mathcal{F}^\oplus$ has jet order $q + 1$. If the nondegenerate leading channel depends on $(D^\oplus)^q \mathcal{F}^\oplus$ with $q \geq 1$, then the leading jet order satisfies $r \geq 2$. By Lemma 3.12, the leading Euler channel has order at least $2r \geq 4$, except for terms whose leading variation is Euler-neutral or degenerate by hypothesis. Hence genuine derivative-curvature leading channels lie outside the admissible second-order structural PDE regime. The density $\|D^\oplus \mathcal{F}^\oplus\|_{L^2}^2$ is the case $q = 1$. $\square$

Proposition 3.15 (Higher Curvature Polynomial Exclusion). *Let P_k be a homogeneous metric-contracted invariant scalar polynomial of degree $k \geq 3$ in the curvature variable, and set*

$$L_k(\omega^\oplus) = P_k(\mathcal{F}^\oplus) dV_h.$$

At background $\omega_0^\oplus$, one has

$$\delta L_k = \langle P'_k(\mathcal{F}_0^\oplus), \delta \mathcal{F}^\oplus \rangle dV_h + \text{terms of lower differential order}, \quad \delta \mathcal{F}^\oplus = D_0^\oplus(\delta \omega^\oplus) + \text{terms of lower differential order}$$

If the polynomial has nondegenerate degree- k leading dependence, the linearized principal symbol carries coefficients depending polynomially on $\mathcal{F}_0^\oplus$, of degree $k - 2$ in the Hessian channel. Hence it depends nontrivially on background amplitude and violates Definition 3.3. Therefore nondegenerate higher curvature polynomial channels of this metric-contracted scalar-density type are excluded from the baseline structural PDE class.

Proof. Differentiate $L_k = P_k(\mathcal{F}^\oplus) dV_h$:

$$\delta L_k = \langle P'_k(\mathcal{F}^\oplus), \delta \mathcal{F}^\oplus \rangle dV_h.$$

At $\omega_0^\oplus$, substitute $\delta \mathcal{F}^\oplus = D_0^\oplus(\delta \omega^\oplus)$ plus terms of lower differential order and integrate by parts to obtain the Euler channel. Linearizing this Euler channel once more gives top-order derivatives acting on $\delta \omega^\oplus$ with coefficient tensor built from the Hessian $P''_k(\mathcal{F}_0^\oplus)$. For a nondegenerate homogeneous degree- k leading polynomial with $k \geq 3$, this coefficient depends nontrivially on $\mathcal{F}_0^\oplus$. Therefore the principal symbol varies with $\omega_0^\oplus$ whenever the relevant background curvature component is nonzero, violating derived field-independent principal-symbol rigidity. $\square$

Lemma 3.16 (Smooth Invariant Generator Reduction). *Every smooth representation-induced curvature scalar on the model fiber*

$$W = \Lambda^2 T_x^* M \otimes \mathfrak{so}(E_x)$$

is a smooth function of finitely many $O(T_x M) \times O(E_x)$ -invariant polynomial generators. In the trace-induced quadratic sector, the metric contraction is the only unoriented generator, while the oriented refinement in dimension $n = 4$ also has the quadratic Chern–Weil generator.

Proof. Compact group $O(T_x M) \times O(E_x)$ acts linearly on W . By Weyl invariant theory, the polynomial invariant algebra is finitely generated by orthogonal contractions [18, 20]. Schwarz’s theorem identifies smooth invariant functions for compact linear group actions as smooth functions of a finite generating set of polynomial invariants [17, 19]. Quadratic trace-induced contraction classification is Theorem 3.40 and Theorem 3.41; the oriented extra generator exists only in dimension four, where it is represented by $\text{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus)$. $\square$

Lemma 3.17 (Constant-Hessian Semilinearity Criterion). *Let $\Phi : W \rightarrow \mathbb{R}$ be a smooth representation-induced invariant curvature scalar. Field-independent semilinearity permits a nonzero second curvature differential only when the non-Euler-neutral part of $d^2 \Phi_F$ is independent of F . In the representation-induced trace class, the only such non-Euler-neutral bilinear form is the metric trace pairing, up to scale. Oriented $n = 4$ Chern–Weil bilinear form is Euler-neutral.*

Proof. At the principal level, admissible curvature variations lie in the exterior-symbol image $\mathcal{C}_\xi(V_{\mathcal{A}_\oplus, x})$, where $\mathcal{C}_\xi(a) = \xi \wedge a$; the Euler projection is applied only after the adjoint exterior-symbol composition. In the second variation of $\Phi(\mathcal{F}^\oplus) dV_h$, the second-order top coefficient contains $d^2 \Phi_F$ applied to principal curvature variations. If the non-Euler-neutral part of this bilinear form varies with F , then the principal coefficient depends on the background curvature amplitude and violates Definition 3.3. Thus semilinearity allows only background-independent non-neutral bilinear forms. By Theorem 3.40 and Theorem 3.41, the representation-induced background-independent quadratic bilinear forms are the metric trace pairing and, in the oriented $n = 4$ refinement, the Chern–Weil pairing. Chern–Weil pairing is Euler-neutral by the Bianchi calculation in Theorem 3.43; metric trace pairing is the Yang–Mills trace pairing. $\square$

Lemma 3.18 (Admissible Principal-Symbol Detectability). *At $x \in M$, let $V_{\mathcal{A}_\oplus, x} \subset T_x^* M \otimes \mathfrak{so}(E_x)$ be the admissible variation fiber and define the admissible curvature-symbol map*

$$\mathcal{C}_\xi : V_{\mathcal{A}_\oplus, x} \longrightarrow \Lambda^2 T_x^* M \otimes \mathfrak{so}(E_x), \quad \mathcal{C}_\xi(a) = \xi \wedge a.$$

Let Q_F be the second curvature differential of a representation-induced curvature scalar at F . If the non-Euler-neutral part of Q_F depends on F and is principal-effective in the sense of Definition 3.9, then there exist a background F_0 , a covector ξ , and admissible variations $a, b \in V_{\mathcal{A}_\oplus, x}$ such that

$$Q_{F_0}(\mathcal{C}_\xi(a), \mathcal{C}_\xi(b))$$

changes with F_0 . If no such admissible principal-symbol directions detect the dependence, then that part contributes no second-order Euler principal coefficient on $\mathcal{A}_\oplus$ and is outside the principal-effective curvature-density category unless it is constant or Euler-neutral.

Proof. Principal part of the curvature variation in the admissible problem is the exterior-symbol image $\mathcal{C}_\xi(a) = \xi \wedge a$. Second-order principal coefficient of the Euler linearization is obtained by restricting the second curvature differential to pairs of such principal curvature variations. Therefore any background dependence of Q_F visible on this restricted image gives a background-dependent

principal coefficient and violates Definition 3.3. Conversely, if the dependence vanishes on all such pairs, it does not enter the second-order Euler principal symbol for admissible variations. By the principal-effectiveness clause in Definition 3.9, such a hidden term is not an admitted non-Euler-neutral curvature-density competitor unless it is constant or Euler-neutral. $\square$

Lemma 3.19 (Constant-Hessian Integration). *Let $\Phi : W \rightarrow \mathbb{R}$ be a smooth representation-induced invariant curvature scalar whose non-Euler-neutral second differential is a fixed trace-class bilinear form, with the oriented $n = 4$ Chern–Weil bilinear form allowed as Euler-neutral. Then*

$$\Phi(F) = \lambda \langle F, F \rangle + \ell(F) + c + \beta P_4(F),$$

where ℓ is linear, c is constant, and $P_4(F)$ is the oriented quadratic Chern–Weil scalar in dimension four and zero otherwise. Invariance under $O(T_x M) \times O(E_x)$ forces $\ell = 0$.

Proof. On the vector space W , a smooth function with constant second differential is a quadratic polynomial plus a linear term and a constant. Fixed non-Euler-neutral bilinear form is the trace-class metric contraction by Lemma 3.17; the only additional admitted quadratic bilinear form is the oriented $n = 4$ Chern–Weil pairing, which is Euler-neutral. A linear invariant would be an $O(T_x M) \times O(E_x)$ -equivariant map $W \rightarrow \mathbb{R}$. Since $W = \Lambda^2 T_x^* M \otimes \mathfrak{so}(E_x)$ has no trivial summand in the metric-only representation, no such nonzero linear invariant exists. Hence $\ell = 0$. $\square$

Theorem 3.20 (Smooth Curvature-Function Semilinearity Exclusion). *Let a density in the principal-effective representation-induced curvature-density regime of Definition 3.9 be locally written as*

$$L_\phi = \phi(P_1(\mathcal{F}^\oplus), \dots, P_r(\mathcal{F}^\oplus)) dV_h,$$

where P_α are homogeneous representation-induced curvature invariants of the assembled curvature. If the Euler–Lagrange operator of L_ϕ has field-independent semilinear second-order principal coefficients on arbitrary curvature backgrounds, then the total non-Euler-neutral second curvature differential of the density is a background-independent bilinear form in the trace-induced quadratic class. Thus, in the trace-induced class, the only non-Euler-neutral surviving term is the metric trace curvature density, up to constant density and oriented $n = 4$ Chern–Weil additions.

Proof. By Lemma 3.16, it is enough to work with a finite set of invariant polynomial generators P_α . Lemma 3.17 shows that the admissible non-Euler-neutral second curvature differential must be a background-independent trace-class bilinear form. Lemma 3.18 gives the corresponding detectability statement on the admissible principal-symbol image. Write $P(\mathcal{F}_0^\oplus) = (P_1(\mathcal{F}_0^\oplus), \dots, P_r(\mathcal{F}_0^\oplus))$. First variation has top differential part

$$\delta L_\phi = \sum_\alpha \frac{\partial \phi}{\partial z_\alpha} (P(\mathcal{F}_0^\oplus)) \delta P_\alpha(\mathcal{F}_0^\oplus; \delta \mathcal{F}^\oplus) dV_h + \text{lower differential order}, \quad \delta \mathcal{F}^\oplus = D_0^\oplus(\delta \omega^\oplus) + \text{lower differential order}$$

Linearizing the Euler channel once more gives second-order top terms whose coefficients contain the Hessian matrix

$$\frac{\partial^2 \phi}{\partial z_\alpha \partial z_\beta} (P(\mathcal{F}_0^\oplus))$$

contracted with the first variations $\delta P_\alpha(\mathcal{F}_0^\oplus; D_0^\oplus \delta \omega^\oplus)$. If this Hessian produces any non-Euler-neutral contribution, the principal coefficients vary with the background curvature $\mathcal{F}_0^\oplus$. This violates Definition 3.3. Hence ϕ is affine in the non-neutral invariant generators, except for affine quadratic terms whose second curvature differential is a constant trace-class bilinear form.

For the remaining affine part, a homogeneous invariant generator of degree at least three has second curvature derivative depending nontrivially on $\mathcal{F}_0^\oplus$, as in Proposition 3.15. Its affine coefficient must therefore vanish unless the corresponding term is Euler-neutral. Lemma 3.19 then identifies the surviving form: the affine constant term is a constant background density, the affine oriented $n = 4$ Chern–Weil term is Euler-neutral by the Bianchi identity, and the remaining non-neutral affine invariant in the trace-induced class is the metric trace curvature density. Thus any smooth invariant not equivalent to the trace quadratic density plus constant and admitted topological terms either has a detected background-dependent principal coefficient, violates the second-order structural regime, or is constant, Euler-neutral, or outside the principal-effective curvature-density category. Nonlinear functions such as $f(\langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle)$ with $f'' \neq 0$ and Born–Infeld-type functions therefore fail the semilinear principal-symbol requirement. $\square$

Theorem 3.21 (Curvature-Source Obstruction Mechanisms). *In the representation-induced curvature-density regime, non-Yang–Mills curvature-density competitors are excluded by the following structural mechanisms:*

- (i) *independent lower-order potentials fail holonomy-obstruction dependence unless they are constant or Euler-neutral (Proposition 3.10);*
- (ii) *derivative-curvature channels fail by differential order (Theorem 3.14);*
- (iii) *nonaffine smooth curvature-invariant functions fail by background-amplitude-dependent principal symbols (Theorem 3.20);*
- (iv) *oriented $n = 4$ Chern–Weil terms are Euler-neutral.*

Proof. Proposition 3.10 gives the first item. Theorem 3.14 gives the derivative-curvature branch. Smooth nonaffine curvature branch is Theorem 3.20, with Proposition 3.15 as the homogeneous-polynomial prototype. Chern–Weil neutrality statement is the variational calculation in Theorem 3.43. $\square$

3.3 First-Jet Complement Exhaustion

First-jet locality leaves more than the assembled curvature. The admissible first-jet fiber splits into the curvature image and two non-curvature complements, the symmetric normal-connection jet and the fully symmetric mixed jet. In the first-jet regime used here, the first complement is removed by gauge freedom. The second is removed by semilinear principal-symbol rigidity. Consequently, curvature dependence is derived for non-Euler-neutral first-jet densities rather than imposed.

Lemma 3.22 (Precise 1-Jet Model). *Let $x \in M$. After choosing an h -orthonormal frame at x and a g_N -orthonormal frame of N_x , the fiber of the 1-jet bundle of admissible connections $J^1\mathcal{C}$ at x identifies with the affine space modeled on*

$$(T_x^*M \otimes \mathfrak{so}(N_x)) \oplus (\text{Sym}^2 T_x^*M \otimes N_x) \oplus (T_x^*M \otimes T_x^*M \otimes \mathfrak{so}(N_x)) \oplus (T_x^*M \otimes \text{Sym}^2 T_x^*M \otimes N_x)$$

with coordinates $(A_i, K_{ij}, \partial_\ell A_i, \partial_\ell K_{ij})$, where $K_{ij} = K_{ji}$.

Proof. Freezing the TT block fixes the Levi–Civita part. Local admissible representatives are (A, K) , where $A \in T^*M \otimes \mathfrak{so}(N)$ is the normal connection block and $K \in \text{Sym}^2 T^*M \otimes N$ is the symmetric mixed tensor. Recording first derivatives at x gives the displayed affine model. $\square$

Lemma 3.23 (Local Jet Surjectivity for Restricted Gauge Action). *Fix $x \in M$. For any prescribed values*

$$u(x) = I, \quad du(x) = 0, \quad \partial_{(\mu}\partial_{\nu)}u(x) = S_{(\mu\nu)} \in \mathfrak{so}(N_x),$$

there exists a local normal gauge transformation realizing these jets.

Proof. Apply Borel finite-jet realization to Lie-algebra valued jets and exponentiate to $\mathrm{SO}(N)$ [15, Thm. 1.2.6]. A cutoff equal to 1 near x preserves the prescribed jet and localizes the gauge representative. $\square$

Lemma 3.24 (Gauge-Killing of Symmetric Normal-Connection Jets). *For any point $x \in M$ and any symmetric tensor $S_{(\mu\nu)} \in \mathfrak{so}(N_x)$, there exists $u \in \mathcal{G}_N$ with $u(x) = I$ and $du(x) = 0$ such that, in any local trivialization,*

$$\partial_{(\mu}\tilde{A}_{\nu)}(x) = \partial_{(\mu}A_{\nu)}(x) + S_{(\mu\nu)},$$

where A is the normal connection block. The transformed connection remains in $\mathcal{A}_{\oplus}$, and the frozen TT Levi-Civita block is unchanged.

Proof. Use the restricted block embedding $\hat{u} = \mathrm{diag}(I_{TM}, u_N)$, so the frozen tangent block is preserved and the mixed block transforms tensorially. Lemma 3.23 supplies a local normal gauge transformation with prescribed symmetric second jet. The connection transformation law then shifts $\partial_{(\mu}A_{\nu)}(x)$ by $S_{(\mu\nu)}$. $\square$

Lemma 3.25 (Frame Reduction on the Orthonormal Bundle). *Fix $x \in M$. Under orthonormal-frame naturality and restricted gauge equivariance, an order-1 natural scalar density is determined by an $O(n) \times O(k)$ -equivariant algebraic map on the 1-jet model at x .*

Proof. Using h -normal coordinates and a local orthonormal trivialization of N at x , the Levi-Civita block vanishes at x , so first jets reduce to algebraic tensors. Let $p \in O(M, h)_x$ and $q \in O(N, g_N)_x$. Naturality under frame changes sends jets in $J_x^1\mathcal{C}$ through the isotropy representation of $O(n) \times O(k)$. A natural scalar density is invariant under this isotropy action. Therefore it is encoded by an $O(n) \times O(k)$ -equivariant map on the algebraic jet model. $\square$

Proposition 3.26 (First-Jet Curvature Map and Complement Decomposition). *Fix $x \in M$ and choose h -normal coordinates and an orthonormal normal frame at x . On the 1-jet fiber, define the linearized curvature map*

$$\mathcal{C}_x : (\partial A, \partial K) \mapsto ((\partial_i A_j - \partial_j A_i), (\partial_\ell K_{ij} - \partial_i K_{\ell j})),$$

with $K_{ij} = K_{ji}$. Normal-connection derivative jets fit into the exact sequence

$$0 \longrightarrow \mathrm{Sym}^2 T_x^* M \otimes \mathfrak{so}(N_x) \longrightarrow T_x^* M \otimes T_x^* M \otimes \mathfrak{so}(N_x) \xrightarrow{\mathrm{Alt}} \Lambda^2 T_x^* M \otimes \mathfrak{so}(N_x) \longrightarrow 0.$$

Mixed derivative jets fit into the exact sequence

$$0 \longrightarrow \mathrm{Sym}^3 T_x^* M \otimes N_x \longrightarrow T_x^* M \otimes \mathrm{Sym}^2 T_x^* M \otimes N_x \xrightarrow{\mathrm{Cod}} \mathcal{W}_{\mathrm{Cod}, x} \longrightarrow 0,$$

where

$$\mathrm{Cod}(T)_{\ell ij} := T_{\ell ij} - T_{i \ell j}$$

and $\mathcal{W}_{\mathrm{Cod}, x} \subset \Lambda^2 T_x^ M \otimes T_x^* M \otimes N_x$ is the Codazzi-type hook component defined by*

$$C_{\ell ij} = -C_{i \ell j}, \quad C_{\ell ij} + C_{ij \ell} + C_{j \ell i} = 0.$$

With the orthogonal splittings induced by h and g_N ,

$$\partial A = \partial A^{\text{curv}} \oplus \partial A^{\text{sym}}, \quad \partial K = \partial K^{\text{curv}} \oplus \partial K^{\text{sym}},$$

where $\partial A^{\text{sym}} \in \text{Sym}^2 T_x^* M \otimes \mathfrak{so}(N_x)$ and $\partial K^{\text{sym}} \in \text{Sym}^3 T_x^* M \otimes N_x$. Its image is exactly the curvature pair $(\text{Alt}(\partial A), \text{Cod}(\partial K))$, and the non-curvature complement is generated by these two kernel components.

Proof. Displayed map is the linearized first-jet part of the (NN) curvature and mixed curvature channels at x in normal coordinates, where background Christoffel symbols vanish at the base point. Normal block curvature is the standard antisymmetrization of a T^*M -valued connection one-form, so its kernel is the symmetric two-tensor slot.

For the mixed block, put $T_{\ell ij} := \partial_\ell K_{ij}$, with $T_{\ell ij} = T_{\ell ji}$. If $\text{Cod}(T) = 0$, then $T_{\ell ij} = T_{i\ell j}$; together with symmetry in the last two indices this makes T fully symmetric. Hence the kernel is $\text{Sym}^3 T_x^* M \otimes N_x$. Conversely, a fully symmetric T is annihilated by Cod . Direct substitution gives the displayed skew and cyclic identities, and these identities characterize the hook image. Orthogonal complete reducibility gives the stated direct-sum splitting. $\square$

Lemma 3.27 (Admissible Mixed-Complement Principal Span). *Let $V = T_x M$ and $W = N_x$. The admissible principal K -variation directions span the fully symmetric mixed complement:*

$$\text{span}\{\text{Sym}(\xi \otimes \kappa) : \xi \in V^*, \kappa \in \text{Sym}^2 V^* \otimes W\} = \text{Sym}^3 V^* \otimes W.$$

Proof. Let (e^i) be a basis of V^* and (ν_α) a basis of W . The tensors $e^i e^j e^\ell \otimes \nu_\alpha$, with $i \leq j \leq \ell$, form a basis of $\text{Sym}^3 V^* \otimes W$. For such a basis tensor choose $\xi = e^i$ and $\kappa = e^j e^\ell \otimes \nu_\alpha$. Then $\text{Sym}(\xi \otimes \kappa)$ is a nonzero scalar multiple of $e^i e^j e^\ell \otimes \nu_\alpha$. Hence every basis tensor lies in the displayed span. $\square$

Lemma 3.28 (Non-Affine Mixed-Complement Hessian Detection). *Let $S \in \text{Sym}^3 T_x^* M \otimes N_x$ denote the fully symmetric mixed complement. Suppose a smooth $O(T_x M) \times O(N_x)$ -invariant representative $\Phi(F, S)$ has genuine non-affine dependence on S , with curvature-image variables F held fixed. Then for some background (F_0, S_0) , the effective complement Hessian*

$$H_S = d_S^2 \Phi_{(F_0, S_0)}$$

is nonzero and is detected by admissible principal K -variation directions.

Proof. Schwarz's theorem for compact linear group actions gives a smooth invariant representative $\Phi(F, S)$ as a smooth function of finitely many invariant polynomial generators [17, 19]. For fixed F , if $d_S^2 \Phi(F, S)$ vanishes on a connected S -domain, then $d_S \Phi(F, S)$ is constant along every line, so $\Phi(F, S)$ is affine in S . Therefore genuine non-affine dependence gives $d_S^2 \Phi_{(F_0, S_0)} \neq 0$ at some background. Lemma 3.27 says admissible principal directions span the full S -space, so a nonzero symmetric bilinear form H_S is nonzero on some pair of admissible directions. $\square$

Proposition 3.29 (Affine Mixed-Complement Residual Exclusion). *In the first-jet admissibility regime, no affine residual*

$$\Phi(F, S) = \ell_F(S) + \Phi_0(F)$$

survives as a genuine non-Euler-neutral complement density.

Proof. A pure affine functional on $S \in \text{Sym}^3 V^* \otimes W$ would require a fixed vector in the standard normal representation. No such vector exists for $k \geq 2$ under the internal normal orthogonal action, and the same conclusion holds under the oriented $\text{SO}(W)$ action. A direct curvature-image coefficient also vanishes: the mixed curvature image has Codazzi hook symmetry, whereas S is fully symmetric, so every direct metric contraction between them is zero; Gauss and normal curvature slots have the wrong tangent or normal parity to give a coefficient in $(\text{Sym}^3 V^* \otimes W)^*$.

Lower-order K -assisted affine contractions can be formed representation-theoretically. A model contraction is

$$S_{abc}^\alpha C_{ade}^\beta K_{bd}^\beta K_{ce}^\alpha.$$

Such a term has second-order curvature/complement principal coefficient depending on the background K . This violates the field-independent semilinear principal-symbol condition in Definition 3.3. An affine branch depending only on lower-order fields and S has no second-order curvature/complement principal coefficient and is degenerate for principal-effectiveness unless it is already a total divergence or an equivalence representative. Hence no affine S -residual remains in the first-jet regime. $\square$

Lemma 3.30 (Normal-Connection Insertion Span). *Assume $k = \dim N_x \geq 2$. Then*

$$\text{span}\{\text{Sym}(a \cdot K_0) : a \in T_x^* M \otimes \mathfrak{so}(N_x), K_0 \in \text{Sym}^2 T_x^* M \otimes N_x\} = \text{Sym}^3 T_x^* M \otimes N_x.$$

Proof. Let (e^i) be a basis of $T_x^* M$ and (ν_α) a basis of N_x . Fix a basis tensor $e^i e^j e^\ell \otimes \nu_\alpha$. Since $k \geq 2$, choose $\beta \neq \alpha$ and a skew endomorphism $E_{\alpha\beta} \in \mathfrak{so}(N_x)$ with $E_{\alpha\beta} \nu_\beta = \nu_\alpha$. Set

$$a = e^i \otimes E_{\alpha\beta}, \quad K_0 = e^j e^\ell \otimes \nu_\beta.$$

Then $\text{Sym}(a \cdot K_0)$ is a nonzero scalar multiple of $e^i e^j e^\ell \otimes \nu_\alpha$. Hence the insertion directions span the full complement. For $k = 1$, $\mathfrak{so}(N_x) = 0$, and this mechanism is absent. $\square$

Proposition 3.31 (Mixed-Complement Hessian Exclusion). *In the first-jet admissibility regime, a principal-effective nonzero total Hessian in the fully symmetric mixed complement*

$$\text{Sym}^3 T_x^* M \otimes N_x$$

violates field-independent semilinear principal-symbol structure.

Proof. Let H_{tot} denote the total effective Hessian in the fully symmetric mixed complement, including pure complement and nonlinear mixed curvature-complement contributions with curvature-image variables held fixed. By Lemma 3.27, admissible K -variation directions

$$\text{Sym}(\xi \otimes \kappa)$$

span the full complement. By Lemma 3.30, normal-connection insertion directions

$$\text{Sym}(a \cdot K_0)$$

also span the full complement when $k \geq 2$.

If $H_{\text{tot}} \neq 0$, some pair of these directions has nonzero H_{tot} -pairing. The corresponding mixed K -to- A top-symbol contribution is

$$H_{\text{tot}}(\text{Sym}(a \cdot K_0), \text{Sym}(\xi \otimes \kappa)),$$

which is linear in K_0 . Curvature-image top symbols in the semilinear class are independent of K_0 . Along a line $K_0(\lambda) = \lambda K_*$, field-independent semilinearity forces the derivative of the total top symbol with respect to λ to vanish. The detected H_{tot} -pairing is precisely this derivative in the complement channel. Hence H_{tot} must vanish on all principal-effective directions. $\square$

Proposition 3.32 (Locality-Only First-Jet Competitor). *Under locality alone, order-1 scalar densities need not factor through $\mathcal{F}^\oplus$ alone. A local gauge-natural density with explicit ∇K dependence is*

$$\mathcal{L}_{\nabla K} := \|D^{h,\perp}K\|_h^2 dV_h.$$

Proof. $D^{h,\perp}K$ is tensorial, $O(n) \times O(k)$ -equivariant, and gauge-natural. Its squared norm against (h, g_N) is a scalar density of first differential order in K . It is not determined by the pointwise value of $\mathcal{F}^\oplus$ alone because two 1-jets with equal $A, \partial A, K$ but different symmetric components of ∂K can have the same $\mathcal{F}^\oplus$ and different $\mathcal{L}_{\nabla K}$. Therefore dependence on $\mathcal{F}^\oplus$ alone does not follow from locality. $\square$

Lemma 3.33 (Mixed Principal-Symbol Identity on the Admissible Symmetric Bundle). *Fix $x \in M$, background (A_0, K_0) , and $\xi \in T_x^*M \setminus \{0\}$. For the mixed channel generated by*

$$\mathcal{L}_{\nabla K} = \|D^{h,\perp}K\|^2,$$

the principal mixed-to-normal symbol on admissible variations

$$\kappa \in \text{Sym}^2 T_x^*M \otimes N_x$$

is the element

$$P_A(\xi, K_0)\kappa \in T_x^*M \otimes \mathfrak{so}(N_x)$$

defined by the trace pairing

$$\langle (P_A(\xi, K_0)\kappa)(X), \eta \rangle_{\mathfrak{so}(N)} = \sum_{r=1}^n \left\langle \kappa(X, e_r), \eta K_0(\xi^\sharp, e_r) \right\rangle_N$$

for every $X \in T_xM$, every $\eta \in \mathfrak{so}(N_x)$, and any h -orthonormal basis $(e_1, \dots, e_n)$ of T_xM . Equivalently,

$$\langle (P_A(\xi, K_0)\kappa)_i, \eta \rangle_{\mathfrak{so}(N)} = \xi^j \langle \kappa_{ir}, \eta K_{0jr} \rangle_N.$$

Proof. Linearize the A -Euler channel

$$E_A = \text{ad}_K^*(D^{h,\perp}K)$$

at (A_0, K_0) . Top-order mixed term in κ is $\text{ad}_{K_0}^*(D_0\kappa)$. At symbol level, replace D_0 by $\xi \otimes (\cdot)$, giving $\text{ad}_{K_0}^*(\xi \otimes \kappa)$. The displayed pairing is the defining adjoint relation for this contraction under the trace pairing on $\mathfrak{so}(N_x)$ and the metric pairing on N_x . Because κ is admissible, $\kappa_{ir} = \kappa_{ri}$, so the contraction is defined on $\text{Sym}^2 T_x^*M \otimes N_x$. $\square$

Theorem 3.34 (Structural Reduced-Symbol Obstruction for the Mixed Channel). *Assume $n \geq 2$ and $k \geq 2$. Fix $x \in M$ and $K_0(x) \neq 0$. Let $\xi \in T_x^*M \setminus \{0\}$ and let P_A be the mixed-to-normal symbol map of Lemma 3.33:*

$$P_A(\xi, K_0)\kappa := \text{ad}_{K_0}^*(\xi \otimes \kappa), \quad \pi_{\text{red}, \xi} : T_x^*M \otimes \mathfrak{so}(N_x) \rightarrow V_{\text{red}, \xi},$$

where

$$V_{\text{red}, \xi} := \{a \in T_x^*M \otimes \mathfrak{so}(N_x) : \iota_{\xi^\sharp} a = 0\}.$$

Assignment

$$(\xi, K_0, \kappa) \longmapsto \pi_{\text{red}, \xi}(P_A(\xi, K_0)\kappa)$$

*is fiberwise canonical and $O(T_xM) \times O(N_x)$ -equivariant. Moreover, for every $K_0(x) \neq 0$, there exist $\xi \neq 0$ and admissible $\kappa \in \text{Sym}^2 T_x^*M \otimes N_x$ such that*

$$\pi_{\text{red}, \xi}(P_A(\xi, K_0)\kappa) \neq 0.$$

Proof. Map

$$\mathcal{P}_{x,\xi,K_0} : \text{Sym}^2 T_x^* M \otimes N_x \rightarrow V_{\text{red},\xi}, \quad \mathcal{P}_{x,\xi,K_0}(\kappa) = \pi_{\text{red},\xi}(P_A(\xi, K_0)\kappa)$$

is a composition of the metric contractions in Lemma 3.33 and orthogonal projection, hence is fiberwise canonical and $O(T_x M) \times O(N_x)$ -equivariant.

It remains to prove nonvanishing. Since $K_0(x) \neq 0$, choose an h -orthonormal basis of $T_x M$ and basis vectors u, z such that

$$v := K_0(u, z) \neq 0.$$

Set $\xi = u^\flat$. Since $n \geq 2$, choose a unit basis vector $s \in T_x M$ with $s \perp u$. Since $k \geq 2$, choose $w \in N_x$ with $\langle w, v \rangle_N = 0$ and $w \neq 0$, and choose $\eta \in \mathfrak{so}(N_x)$ such that $\eta v = w$; such an η is the skew rotation in the plane spanned by v and w , extended by zero on its orthogonal complement. Define an admissible symmetric variation $\kappa \in \text{Sym}^2 T_x^* M \otimes N_x$ by requiring

$$\kappa(s, z) = \kappa(z, s) = w$$

and setting all components orthogonal to the symmetric span of $s \odot z$ to zero.

Evaluate the defining pairing of Lemma 3.33 at the transverse vector s and the skew endomorphism η . In the chosen orthonormal frame, the only contributing component is the z -slot, and hence

$$\langle (P_A(\xi, K_0)\kappa)(s), \eta \rangle_{\mathfrak{so}(N)} = \langle w, \eta v \rangle_N = |w|_N^2 \neq 0.$$

Thus $(P_A(\xi, K_0)\kappa)(s) \neq 0$. Since $s \perp \xi^\sharp$, this component lies in the transverse normal one-form block and survives the projection $\pi_{\text{red},\xi}$. Therefore

$$\pi_{\text{red},\xi}(P_A(\xi, K_0)\kappa) \neq 0.$$

□

Proposition 3.35 (Explicit Principal-Symbol Rigidity Exclusion for $\|D^{h,\perp}K\|^2$ Channel). *Assume $n \geq 2$ and $k \geq 2$. Consider*

$$\mathcal{L}_{\nabla K} = \|D^{h,\perp}K\|^2,$$

with coupled variables (A, K) , where $D^{h,\perp}$ contains the normal connection A . Its Euler channels are

$$E_K = (D^{h,\perp})^\dagger D^{h,\perp} K, \quad E_A = \text{ad}_K^*(D^{h,\perp} K).$$

Linearize around a background (A_0, K_0) with $K_0 \neq 0$, using admissible variations $\kappa \in \text{Sym}^2 T^ M \otimes N$:*

$$\delta E_A = \text{ad}_{K_0}^*(D_0 \kappa) + \text{ad}_\kappa^*(D_0 K_0) + \text{terms of lower differential order}.$$

Hence the highest-order channel in the A -equation for κ has symbol

$$\sigma_\xi(\delta E_A) \supset \text{ad}_{K_0}^*(\xi \otimes \kappa),$$

whose coefficient depends on K_0 . Therefore the principal propagation channel is field-amplitude dependent and violates field-independent principal-symbol rigidity. Because the coefficient tensor in the principal symbol contains K_0 , the highest-order operator is quasilinear rather than semilinear in the sense of Definition 3.3; this failure persists after gauge reduction by Theorem 3.34.

Proof. From $\mathcal{L}_{\nabla K}$, variation in K (with $\kappa \in \text{Sym}^2 T^*M \otimes N$) gives the rough Laplace channel $(D^{h,\perp})^\dagger D^{h,\perp} K$. Variation in A enters through the connection part of $D^{h,\perp}$:

$$\delta_A(D^{h,\perp} K) = a \cdot K, \quad \delta_K(D^{h,\perp} K) = D^{h,\perp} \kappa.$$

After integration by parts, the A -Euler channel is $\text{ad}_K^*(D^{h,\perp} K)$. Linearizing at (A_0, K_0) gives the displayed expression. Taking principal symbol in the κ -channel yields $\text{ad}_{K_0}^*(\xi \otimes \kappa)$, explicitly K_0 -dependent. This is exactly the field-independent principal-symbol failure. By Lemma 3.33, the top-order mixed-to-normal contribution on admissible symmetric κ is the trace-pairing adjoint $P_A(\xi, K_0)\kappa$, characterized by

$$\langle (P_A(\xi, K_0)\kappa)(X), \eta \rangle_{\mathfrak{so}(N)} = \sum_{r=1}^n \left\langle \kappa(X, e_r), \eta K_0(\xi^\sharp, e_r) \right\rangle_N.$$

Thus the leading coefficient tensor depends on K_0 . Hence the operator is quasilinear at top order in this channel, whereas the structural regime requires a field-independent (semilinear) principal symbol. For reduced-symbol verification, write the mixed/normal principal block as

$$\sigma_\xi(\delta E) \sim \begin{pmatrix} |\xi|_h^2 \text{Id}_A & P_A(\xi, K_0) \\ 0 & |\xi|_h^2 \text{Id}_K \end{pmatrix}, \quad P_A(\xi, K_0)\kappa = \text{ad}_{K_0}^*(\xi \otimes \kappa).$$

Restricted gauge has principal symbol $\sigma_\xi(d)\epsilon = (\xi \otimes \epsilon, 0)$, so gauge reduction removes only the longitudinal normal block in A . Theorem 3.34 gives explicit (ξ, κ) with nonzero transverse projection $\pi_{\text{red}, \xi} \circ P_A(\xi, K_0)\kappa$ in the regime $n \geq 2, k \geq 2$. Therefore the K_0 -dependent top-order coefficient survives reduced-symbol quotienting. $\square$

Proposition 3.36 (General Semilinearity of the Projected Yang–Mills Channel). *At any admissible background $\omega^\oplus \in \mathcal{A}_\oplus$, the top-order symbol of the linearized projected Yang–Mills operator*

$$\delta(\Pi_{\mathcal{A}_\oplus}(D^\oplus)^\dagger \mathcal{F}^\oplus)$$

is the fixed exterior-symbol composition

$$a \longmapsto \Pi_{\mathcal{A}_\oplus} \iota_{\xi^\sharp}(\xi \wedge a)$$

on admissible variations. It is independent of the connection amplitude. Raw, quotient, and Hodge–Laplace decompositions recorded later are the special block-diagonal-background calculations.

Proof. Curvature variation has top-order part $\delta \mathcal{F}^\oplus = D^\oplus a$, whose principal symbol is $\xi \wedge a$. Formal adjoint contributes the fixed contraction $\iota_{\xi^\sharp}$ at principal level. Connection coefficients in $D^\oplus$ and $(D^\oplus)^\dagger$ enter only lower-order terms and therefore do not affect the principal symbol. Admissible projection $\Pi_{\mathcal{A}_\oplus}$ is fixed by the frozen tangent block and the symmetric mixed projection, hence is also independent of the background amplitude. Thus the second-order top symbol is the displayed fixed composition at every admissible background. Detailed kernel, quotient, and auxiliary Hodge–Laplace statements require the block-diagonal assumptions stated in Section 6 and Appendix D. $\square$

Remark 3.37 (Semilinearity of the Yang–Mills Channel). *Yang–Mills curvature satisfies the semilinear principal-symbol condition because the linearized top-order raw symbol is the fixed exterior-symbol composition on the auxiliary B -model and the fixed projected exterior-symbol composition on the admissible mixed block. Its Hodge–Laplace representative has scalar symbol $|\xi|_h^2 \text{Id}$, but that scalar representative is not the raw ungauged Hessian.*

Proposition 3.38 (Channel-by-Channel Obstruction Mechanisms). *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2$, $k \geq 2$, the first-jet complement channels from Proposition 3.26 have the following obstruction mechanisms:*

- ∂A^{sym} is gauge-removable by Lemma 3.24 and does not define an independent gauge-invariant leading channel.
- ∂K^{sym} admits mixed-derivative prototype channels represented by $\|D^{h,\perp} K\|^2$, and Proposition 3.35 gives background-amplitude-dependent principal coefficients together with nonvanishing reduced projected obstruction by Theorem 3.34.
- derivative-curvature channels, for example $\|D^\oplus \mathcal{F}^\oplus\|^2$, are excluded by order via Theorem 3.14.
- nonaffine smooth curvature-invariant functions are excluded by background-dependent principal symbols via Theorem 3.20, with higher curvature polynomials as the prototype in Proposition 3.15.

Proof. Lemma 3.24 gives the first item. Proposition 3.35 together with Theorem 3.34 gives the second item in the regime $n \geq 2$, $k \geq 2$. Theorem 3.14 and Theorem 3.20 give the third and fourth items. The full smooth complement classification is Theorem 3.39. $\square$

Theorem 3.39 (First-Jet Complement Exhaustion). *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2$, $k \geq 2$, every density in the first-jet admissibility regime of Definition 3.7 has no genuine non-Euler-neutral dependence on the first-jet complements in Proposition 3.26. Equivalently, modulo the stated equivalence relation, its non-Euler-neutral first-jet dependence factors through the assembled curvature $\mathcal{F}^\oplus$.*

Proof. Theorem 3.11 gives finite-jet dependence, and Theorem 3.13 reduces nondegenerate top-order channels to first jets. Lemma 3.22 identifies the admissible jet coordinates $(A, K, \partial A, \partial K)$, and Lemma 3.25 reduces order-one natural scalar densities to $O(n) \times O(k)$ -equivariant algebraic functions on that fiber; see also [21, 8, 23]. Compact invariant theory and Schwarz’s theorem identify smooth invariant functions with smooth functions of polynomial generators [18, 17, 19, 20].

Proposition 3.26 splits the first derivative coordinates into the curvature image and two complements. The symmetric normal-connection complement is gauge-removable by Lemma 3.24; gauge-natural scalar densities therefore factor through the quotient by this slot.

Let $S \in \text{Sym}^3 T_x^* M \otimes N_x$ denote the fully symmetric mixed complement. If a representative has genuine non-affine dependence on S , Lemma 3.28 gives a nonzero effective complement Hessian detected by admissible principal K -variations. The connection term in $D^{h,\perp} K$ gives the mixed K -to- A top-symbol contribution

$$H_S(\text{Sym}(a \cdot K_0), \text{Sym}(\xi \otimes \kappa)).$$

Proposition 3.31 excludes this channel in the semilinear principal-symbol regime. The same proposition handles cancellations among several pure complement and nonlinear mixed curvature-complement generators through the total effective complement Hessian H_{tot} .

If $H_{\text{tot}} = 0$, Lemma 3.27 leaves no hidden Hessian-invisible subspace. The S -dependence is affine, degenerate, or equivalent. Proposition 3.29 removes affine residuals in the first-jet regime. Constants, total divergences, Euler-neutral additions, degenerate leading dependence, and terms already represented by functions of $\mathcal{F}^\oplus$ are handled by the equivalence relation in Definition 3.7. No genuine non-Euler-neutral complement dependence remains. $\square$

3.4 Quadratic Curvature Density

The exclusion mechanisms leave a single non-Euler-neutral curvature density in the representation-induced class. Algebraic invariant theory identifies it as the trace metric contraction of $\mathcal{F}^\oplus$. The oriented four-dimensional Chern–Weil contraction is allowed only as an Euler-neutral addition.

Theorem 3.40 (Algebraic Quadratic Invariant Space). *Fix $x \in M$ and set $W_x = \Lambda^2 T_x^* M \otimes \mathfrak{so}(m)$. In the representation-induced curvature-density regime, the curvature is treated as a full $\mathfrak{so}(E_x)$ -valued two-form, with Lie-algebra pairing induced from the defining orthogonal representation. Under the unoriented $O(n) \times O(m)$ algebraic action,*

$$\left(\text{Sym}^2(W_x^*) \right)^{O(n) \times O(m)} = \text{span}\{\mathcal{I}_{\text{met}}\}.$$

If an orientation of $T_x M$ is fixed and the tangent-frame symmetry is reduced to $SO(n)$, while retaining the representation-induced $O(m)$ -naturalness in the $\mathfrak{so}(m)$ factor, then the oriented refinement is

$$\left(\text{Sym}^2(W_x^*) \right)^{SO(n) \times O(m)} = \begin{cases} \text{span}\{\mathcal{I}_{\text{met}}\}, & n \neq 4, \\ \text{span}\{\mathcal{I}_{\text{met}}, \mathcal{I}_{\text{top}}\}, & n = 4, \end{cases}$$

where

$$\mathcal{I}_{\text{met}}(F, F) = \langle F, F \rangle_h, \quad \mathcal{I}_{\text{top}}(F, F) = \langle F, \star_h F \rangle_h.$$

Proof. Any quadratic scalar is the composition of:

- (i) an invariant contraction on two copies of $\Lambda^2 T_x^* M$, and
- (ii) an $O(m)$ -natural representation-induced symmetric bilinear form on $\mathfrak{so}(m)$.

By Lemma 2.6 and Theorem 2.10, the Lie-algebra part is fixed up to scale by the trace pairing class. For the form-index part, the $O(n)$ -invariant bilinear forms on Λ^2 are generated only by

$$(\alpha, \beta) \mapsto \langle \alpha, \beta \rangle_h.$$

After an orientation is fixed, the $SO(n)$ -invariant bilinear forms on Λ^2 also allow

$$(\alpha, \beta) \mapsto \langle \alpha, \star_h \beta \rangle_h.$$

This second contraction exists only when $\star_h : \Lambda^2 \rightarrow \Lambda^{n-2}$ preserves degree, hence only for $n = 4$. Therefore the unoriented invariant quadratic space is always one-dimensional, while the oriented refinement has the stated additional $n = 4$ generator. Equivalently, the ε -contraction requires a fixed oriented volume form. $\square$

Proposition 3.41 (Index Expansion of Quadratic Contractions). *Let (e_i) be an h -orthonormal basis of $T_x M$, and write*

$$F = \frac{1}{2} F_{ij}^a e^i \wedge e^j \otimes T_a, \quad F_{ij}^a = -F_{ji}^a,$$

for an orthonormal basis (T_a) of $\mathfrak{so}(m)$ in a chosen representative of the trace pairing class. Any $O(n) \times O(m)$ -invariant quadratic scalar in the representation-induced class is a linear combination of:

$$Q_1(F) = \sum_{i,j,a} F_{ij}^a F_{ij}^a.$$

If an orientation of $T_x M$ is fixed and $n = 4$, the oriented $SO(n) \times O(m)$ -invariant refinement has one additional generator,

$$Q_2(F) = \sum_{i,j,k,\ell,a} \varepsilon_{ijkl} F_{ij}^a F_{k\ell}^a.$$

No other independent quadratic contraction exists.

Proof. This is the index-level expansion of Theorem 3.40: unoriented orthogonal invariance reduces Lie-algebra contractions to δ_{ab} in the trace class and tangent-index contractions to δ -channels. Dimension-four ε -channel belongs only to the oriented $SO(n)$ -refinement. $\square$

Definition 3.42 (Lagrangian Equivalence). *Two admissible Lagrangians L_1, L_2 are equivalent, written $L_1 \sim L_2$, if*

$$L_2 = \alpha L_1 + d\Theta + c dV_h + \beta \mathcal{P}_4(\mathcal{F}^\oplus)$$

for some nonzero constant $\alpha \in \mathbb{R}$, local $(n-1)$ -form Θ , constants $c, \beta \in \mathbb{R}$, and

$$\mathcal{P}_4(\mathcal{F}^\oplus) = \begin{cases} \text{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus), & n = 4 \text{ in the oriented refinement,} \\ 0, & \text{otherwise.} \end{cases}$$

Here $c dV_h$ is the constant background density. Equivalently, they differ by global multiplicative scale, total divergence, constant background density, and the oriented Euler-neutral Chern–Weil density in dimension four.

Theorem 3.43 (Representation-Induced Curvature Density Classification). *Let M^n be an oriented Riemannian manifold, and let $E = TM \oplus N$ have rank $m = n + k$. Under Definition 3.9, the unique non-Euler-neutral local scalar density in the representation-induced curvature-sourced class is, up to the equivalence relation of Definition 3.42,*

$$\text{tr}(\mathcal{F}^\oplus \wedge \star_h \mathcal{F}^\oplus).$$

In oriented dimension $n = 4$, the additional quadratic invariant $\text{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus)$ is Euler-neutral in the boundary regime because its first variation is exact by the Bianchi identity. It contributes no bulk term to the Euler–Lagrange field equations. Higher Chern–Weil forms are outside the curvature-density regime, and the statement is not a classification of all scalar densities invariant under the weaker frozen-split group alone.

Proof. By Theorem 3.20, semilinearity forces the non-Euler-neutral curvature-invariant dependence to be affine and removes affine generators whose second curvature derivative is background-dependent. Only the quadratic non-neutral trace-induced generator remains. By Theorem 3.40 and Theorem 3.41, the affine quadratic local invariants in the representation-induced trace class are linear combinations of the metric contraction and, only in dimension $n = 4$, the topological wedge contraction. This is the standard orthogonal invariant-theory/Chern–Weil classification at quadratic order; see [18, 19, 1].

Metric contraction gives the Yang–Mills density $\text{tr}(\mathcal{F}^\oplus \wedge \star_h \mathcal{F}^\oplus)$. In $n = 4$, the second generator is $\text{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus)$. Equivalently, under a chosen normalization of the trace pairing class and Hodge metric, $\text{tr}(\mathcal{F}^\oplus \wedge \star_h \mathcal{F}^\oplus) = \langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle dV_h$.

For variational neutrality of the topological term, use $\delta \mathcal{F}^\oplus = D^\oplus(\delta \omega^\oplus)$:

$$\delta \text{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus) = 2 \text{tr}(D^\oplus \delta \omega^\oplus \wedge \mathcal{F}^\oplus) = 2 d \text{tr}(\delta \omega^\oplus \wedge \mathcal{F}^\oplus) - 2 \text{tr}(\delta \omega^\oplus \wedge D^\oplus \mathcal{F}^\oplus).$$

Bianchi identity $D^\oplus \mathcal{F}^\oplus = 0$ removes the bulk term. Hence

$$\delta \int_M \text{tr}(\mathcal{F}^\oplus \wedge \mathcal{F}^\oplus) = 2 \int_{\partial M} \text{tr}(\delta \omega^\oplus \wedge \mathcal{F}^\oplus) = 0$$

under the boundary regime: either $\partial M = \emptyset$ (closed manifold) or variations are compactly supported, so boundary terms vanish. Therefore topological additions do not modify the Euler–Lagrange operator and are included in Definition 3.42. The metric trace density is consequently the unique non-Euler-neutral representative in the stated regime. $\square$

Definition 3.44 (Curvature Energy Functional). *Trace-class curvature energy is*

$$V(\omega^\oplus) = \frac{1}{2g^2} \int_M \langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle dV_h$$

on the admissible affine configuration space $\mathcal{A}_\oplus$.

Definition 3.45 (Admissible Kinetic Extension). *An admissible kinetic extension on $\mathcal{A}_\oplus$ consists of a covariant scalar density in the representation-induced curvature-density regime of Definition 3.9 together with the ambient trace-induced ultralocal quadratic kinetic pairing on admissible velocity fibers.*

Proposition 3.46 (Trace-Induced Pairing Fixes the Velocity Hessian). *For admissible kinetic extensions on $\mathcal{A}_\oplus$ in the sense of Definition 3.45, the velocity Hessian is fixed by the same representation-induced invariant pairing that defines the curvature energy channel.*

Proof. An admissible kinetic Lagrangian has a fiberwise velocity Hessian on $T_\omega \mathcal{A}_\oplus$. Gauge-naturality alone permits separate invariant scales on non-isomorphic irreducible summands. Definition 3.45 imposes the stronger ambient trace-induced pairing. By Definition 2.7 and Proposition 2.11, the orthogonal representation induces a unique invariant trace class up to scale. Therefore the kinetic Hessian is fixed within that trace-induced class, and the kinetic metric and curvature energy are generated by the same pairing. $\square$

4 Euler Equation of the Curvature Density

By Theorem 3.43, the admissible non-Euler-neutral variational density is represented by

$$L_{\text{YM}}(\omega^\oplus) := \frac{1}{2} \text{tr}(\mathcal{F}^\oplus \wedge \star_h \mathcal{F}^\oplus) = \frac{1}{2} \langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle dV_h.$$

Its first variation gives the ambient Yang–Mills Euler expression before projection to the admissible variation bundle [24].

Proposition 4.1 (Ambient and Admissible Euler Operators). *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2$, $k \geq 2$, the first variation identity is*

$$\delta \int_M L_{\text{YM}} = \int_M \langle \delta \omega^\oplus, (D^\oplus)^\dagger \mathcal{F}^\oplus \rangle dV_h + \int_{\partial M} \text{tr}(\delta \omega^\oplus \wedge \star_h \mathcal{F}^\oplus).$$

Under the boundary regime fixed in Section 2, the boundary term vanishes. The ambient Euler expression is therefore

$$\mathcal{E}_{\text{amb}} = - \star_h D^\oplus (\star_h \mathcal{F}^\oplus),$$

equivalently $(D^\oplus)^\dagger \mathcal{F}^\oplus$ under the adjoint convention. On the restricted configuration space $\mathcal{A}_\oplus$, stationarity is the projected Yang–Mills equation

$$\mathcal{E}_{\text{YM}} = \Pi_{\mathcal{A}_\oplus} \mathcal{E}_{\text{amb}} = \Pi_{\mathcal{A}_\oplus} ((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0.$$

Equivalently, the equation is the normal block and symmetric mixed projection of the ambient Yang–Mills expression; it does not impose the frozen TT block or the skew mixed block.

Proof. Curvature variation gives $\delta \mathcal{F}^\oplus = D^\oplus(\delta \omega^\oplus)$. Compatibility of the trace pairing with $D^\oplus$ gives

$$\delta \int_M L_{\text{YM}} = \int_M \text{tr}(D^\oplus(\delta \omega^\oplus) \wedge \star_h \mathcal{F}^\oplus).$$

Covariant integration by parts gives the stated bulk term and boundary current. Ambient stationarity would impose the full Yang–Mills expression. In the present variational problem, $\delta \omega^\oplus$ ranges only over $T_{\omega^\oplus} \mathcal{A}_\oplus$. Hence the Euler–Lagrange operator is the L^2 -orthogonal projection of the ambient expression onto the admissible variation bundle, namely $\Pi_{\mathcal{A}_\oplus} ((D^\oplus)^\dagger \mathcal{F}^\oplus)$. Appendix C gives the projected block form. $\square$

5 Covariant Identity and Gauge Constraint

Restricted normal gauge symmetry gives an off-shell identity for the projected Euler operator. This identity is differential and covariant; the canonical Gauss constraint appears only after the fixed-product time split of Section 8.

Proposition 5.1 (Internal Gauge Noether Identity). *For $\epsilon \in \Gamma(\mathfrak{so}(N))$, the infinitesimal restricted gauge action has admissible variation blocks*

$$\delta_\epsilon A = D_A \epsilon, \quad \delta_\epsilon K = \epsilon \cdot K.$$

Embed ϵ as the normal block of an $\mathfrak{so}(E)$ -valued section. Then $\delta_\epsilon \omega^\oplus = D^\oplus \epsilon$ lies in $T_{\omega^\oplus} \mathcal{A}_\oplus$. Pairing the projected Euler operator $\mathcal{E}_{\text{YM}} = \Pi_{\mathcal{A}_\oplus} \mathcal{E}_{\text{amb}}$ with this generator gives the normal-block adjoint identity through the canonical inclusion $T_{\omega^\oplus} \mathcal{A}_\oplus \subset \Omega^1(M, \mathfrak{so}(E))$. The restricted normal gauge symmetry of the classified Lagrangian yields

$$\pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_{\text{YM}}) = 0.$$

Proof. Embed $\epsilon \in \Gamma(\mathfrak{so}(N))$ as the normal block of an $\mathfrak{so}(E)$ -valued section. Infinitesimal action of $\text{diag}(I_{TM}, u)$ fixes the tangent block, sends the normal connection variation to $D_A \epsilon$, and sends the mixed tensor variation to $\epsilon \cdot K$. Since this action is only on the normal index, symmetry of K in the tangent slots is preserved; hence $D^\oplus \epsilon \in T_{\omega^\oplus} \mathcal{A}_\oplus$. Gauge invariance of the classified density gives $\langle \mathcal{E}_{\text{YM}}, D^\oplus \epsilon \rangle_{L^2} = 0$ for all ϵ . Covariant integration by parts, as recorded in Appendix B, gives $\langle \pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_{\text{YM}}), \epsilon \rangle_{L^2} = 0$, hence $\pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_{\text{YM}}) = 0$. $\square$

Remark 5.2 (Covariant Identity and Canonical Gauss Constraint). *The covariant identity*

$$\pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_{\text{YM}}) = 0$$

is an off-shell consequence of restricted internal gauge symmetry. The canonical Gauss constraint is obtained later from the fixed-product split, where the temporal normal connection A_0 becomes a multiplier.

Remark 5.3 (Curvature–Momentum Module). *For the classified Yang–Mills channel, the ambient Euler expression lies in the differential module generated by curvature and its covariant momenta:*

$$\mathcal{E}_{\text{amb}} \in \mathcal{I}_{\text{curv}}.$$

Here $\mathcal{E}_{\text{YM}} = \Pi_{\mathcal{A} \oplus} \mathcal{E}_{\text{amb}}$. Proposition E.5 proves the containment in Appendix E. The containment records the algebraic closure of the Euler expression under the curvature source; the classification itself is already completed by Theorem 3.43.

6 Linearization and Symbol Structure

Linearization is recorded only at a block-diagonal compatible background,

$$K_0 = 0, \quad F_0^\perp = 0.$$

At such a background the mixed compatibility block is absent and the normal curvature has no principal contribution, so the leading symbol separates into normal and mixed pieces. At this background the principal calculation separates three objects: the raw projected Euler symbol, the normal quotient by internal gauge, and the auxiliary Hodge–Laplace representative. The last object is elliptic; the raw mixed Hessian is not asserted to be elliptic. The auxiliary Hodge–Laplace representative is obtained by adding the complementary codifferential-symbol composition to the raw exterior-symbol composition; it is used only to record the elliptic representative, not to replace the raw Euler Hessian.

Proposition 6.1 (Raw, Quotient, and Auxiliary Symbol Separation). *At a block-diagonal compatible background with $K_0 = 0$ and $F_0^\perp = 0$, let an admissible variation be written*

$$a = (a_\perp, a_K),$$

where $a_\perp$ is the normal-connection variation and a_K is the admissible mixed variation. For $\xi \neq 0$, the projected raw Euler linearization has principal symbol

$$\sigma_\xi^{\text{raw}}(a) = \left(|\xi|_h^2 a_\perp - \xi \otimes \iota_{\xi^\#} a_\perp, \Pi_{\text{sym}}(|\xi|_h^2 a_K - \xi \otimes \iota_{\xi^\#} a_K) \right).$$

The kernel of the mixed raw block is computed in Appendix D. Internal gauge-symbol image is $(\xi \otimes \epsilon, 0)$, hence only the normal longitudinal kernel is internal gauge. On the normal quotient, the raw symbol is the standard Yang–Mills transverse symbol. Projected mixed raw block has double-longitudinal symmetric kernel $\xi \otimes \xi \otimes N_x$, not generated by internal gauge. Its auxiliary Hodge–Laplace representative has symbol

$$\sigma_\xi^\Delta(a) = |\xi|_h^2(a_\perp, a_K).$$

Thus ellipticity of the mixed block is a statement about the auxiliary Hodge–Laplace representative, not about the raw ungauged Euler Hessian.

Proof. At the stated background, the connection is block-diagonal at principal level and the variation of curvature is

$$\delta \mathcal{F}^\oplus = D_0^\oplus a$$

modulo lower-order commutator terms. Hence

$$\sigma_\xi(D_0^\oplus)(a) = \xi \wedge a, \quad \sigma_\xi((D_0^\oplus)^\dagger)(b) = \iota_{\xi^\#} b.$$

The raw principal symbol of the ambient composition is therefore

$$a \longmapsto \iota_{\xi^\#}(\xi \wedge a) = |\xi|_h^2 a - \xi \otimes \iota_{\xi^\#} a.$$

Restricting to the normal block gives the first component in the displayed identity. On the mixed block, admissibility imposes $a_K \in \text{Sym}^2 T_x^* M \otimes N_x$, and the Euler projection applies Π_{sym} after the exterior-symbol composition on the unconstrained mixed model. This gives the second component.

The restricted internal gauge symbol is $\epsilon \mapsto (\xi \otimes \epsilon, 0)$, so it removes only the normal longitudinal kernel. The normal quotient is therefore the usual transverse Yang–Mills symbol. Lemma D.1 in Appendix D gives the mixed raw kernel $\xi \otimes \xi \otimes N_x$, which is not generated by internal gauge. Adding the complementary codifferential-symbol composition $\xi \wedge \iota_{\xi^\#}$ gives the auxiliary Hodge–Laplace representative with scalar symbol $|\xi|_h^2 \text{Id}$. Thus ellipticity of the mixed block belongs only to the auxiliary representative, not to the raw ungauged Euler Hessian. $\square$

7 Higher-Rank Field-Law Classification

Remark 7.1 (Single Pairing Controls Configuration Metric and Curvature Density). *A single representation-induced Ad-invariant trace pairing on $\mathfrak{so}(E)$, fixed up to scale, supplies both the kinetic L^2 metric on $T\mathcal{A}_\oplus$ and the normalization class of the trace quadratic curvature density. Thus the same trace pairing fixes the configuration-space metric and the curvature-density normalization after the first-jet theorem has reduced the non-Euler-neutral density to a curvature representative.*

Theorem 3.39 supplies curvature dependence in the first-jet regime. Definition 3.9 then gives the trace-induced curvature representative class for the tangent-normal connection $\nabla^\oplus$.

Theorem 7.2 (Higher-Rank First-Jet Classification). *Assume:*

1. (M, h) is a smooth Riemannian manifold in the boundary regime, with $n = \dim M \geq 2$.
2. $E = TM \oplus N$ is an orthogonal splitting with fixed background metric data (h, g_N) .
3. the normal rank satisfies $k = \text{rank}(N) \geq 2$.
4. the tangent Levi–Civita block is frozen and the configuration space is $\mathcal{A}_\oplus$.
5. the variational density belongs to the first-jet admissibility regime of Definition 3.7; after the complement-exhaustion reduction of Theorem 3.39, its curvature representative lies in the representation-induced curvature-density regime of Definition 3.9.

Under these conditions, and modulo the equivalence relation of Definition 3.42:

- (i) the unique non-Euler-neutral representative is the trace quadratic curvature density

$$\mathcal{L}_{\text{YM}}(\omega^\oplus) = \frac{1}{2g^2} \langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle dV_h;$$

- (ii) the Euler–Lagrange equation on the restricted configuration space $\mathcal{A}_\oplus$ is

$$\Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0;$$

- (iii) the restricted normal gauge symmetry gives the off-shell identity

$$\pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_{\text{YM}}) = 0;$$

(iv) at a block-diagonal compatible background with $K_0 = 0$ and $F_0^\perp = 0$, the linearization has the raw, normal-quotient, and auxiliary Hodge–Laplace symbol structure of Proposition 6.1.

No ellipticity statement is asserted for the raw mixed Hessian without the auxiliary representative.

Proof. Theorem 3.39 removes genuine non-Euler-neutral dependence on the non-curvature first-jet complements, so the density has a curvature representative. Theorems 3.14 and 3.20 remove derivative-curvature and nonaffine curvature channels in the second-order semilinear regime. Theorem 3.43 then leaves the trace quadratic density as the unique non-Euler-neutral representative, with the oriented $n = 4$ Chern–Weil term absorbed by Definition 3.42. Remark 2.12 fixes the normalization of the trace pairing, giving the displayed g^{-2} normalization.

Section 4 gives the projected Euler equation for this representative. Section 5 gives the internal gauge Noether identity. Section 6 gives the raw, quotient, and auxiliary symbol separation at the stated block-diagonal compatible background. This proves the density classification, the projected field equation, the internal gauge identity, and the stated symbol separation. $\square$

8 Canonical Gauge Interface

The uniqueness theorem gives the covariant higher-rank field equation. The fixed-product split below records only the classical phase-space interface of that equation. The fixed-product Legendre split uses the standard geometric Hamiltonian viewpoint on field theory [11, 12, 13, 14]. No ADM identification, Lorentzian Cauchy theorem, quantization, or global symplectic reduction is asserted in the higher-rank sector.

Assumption 8.1 (Formal Fixed-Product Canonical Regime). *Canonical constructions are taken on a fixed product*

$$M \cong \mathbb{R} \times \Sigma, \quad h = dt^2 + \gamma, \quad dV_h = dt \wedge dV_\gamma,$$

with unit lapse and zero shift. The fields are smooth, Σ is compact without boundary or boundary terms vanish under the imposed decay conditions, and the following data are fixed.

1. *the tangent metric h , normal metric g_N , tangent Levi–Civita block, and product split;*
2. *the internal gauge group $\mathcal{G}_N = \Gamma(\mathrm{SO}(N))$, with no diffeomorphism quotient;*
3. *the adapted temporal-mixed sector*

$$K(\partial_t, \cdot) = 0.$$

Equivalently $K_{00} = 0$ and $K_{0i} = 0$, with variations preserving these conditions. Phase-space constructions are formal on smooth admissible fields; no Banach, Fréchet, global quotient-manifold, foliation-covariant, ADM, or Lorentzian evolution structure is asserted.

Positive trace normalization is fixed as in Remark 2.12. Common global factors in the curvature density are suppressed. Restoring them rescales momenta and Hamiltonian without changing the canonical structure. Since $h = dt^2 + \gamma$ is Riemannian, the Legendre transform is the formal Riemannian Legendre transform; the magnetic term enters the Hamiltonian with the corresponding sign.

Proposition 8.2 (Adapted Field Decomposition). *Under Assumption 8.1, the normal connection and mixed field decompose as*

$$\omega^\perp = A_0 dt + A, \quad A \in \Omega^1(\Sigma, \mathfrak{so}(N)),$$

and

$$K = K_\Sigma, \quad K_\Sigma \in \Gamma(\text{Sym}^2 T^* \Sigma \otimes N).$$

Assembled connection splits as

$$\omega^\oplus = \omega_0^\oplus dt + \omega_\Sigma^\oplus, \quad \omega_0^\oplus = \omega_0^h \oplus A_0, \quad \omega_\Sigma^\oplus = \omega_\Sigma^h \oplus A + \Psi(K_\Sigma).$$

Formal phase variables are

$$(A, K_\Sigma),$$

while A_0 is a multiplier field. Tangent Levi-Civita block and product split are fixed background data.

Writing K for K_Σ , the electric and magnetic curvature channels are

$$E_i^\oplus = (\mathcal{F}^\oplus)_{0i}, \quad B_{ij}^\oplus = (\mathcal{F}^\oplus)_{ij}.$$

Their normal and mixed electric projections are

$$E_\perp = \dot{A} - D_A A_0, \quad E_K = D_t^{A_0} K, \quad D_t^{A_0} K = \partial_t K + A_0 \cdot K.$$

Their magnetic projections are

$$B_T = (G)_{\Sigma\Sigma}, \quad B_K = (C)_{\Sigma\Sigma}, \quad B_\perp = (S)_{\Sigma\Sigma}.$$

Consequently,

$$\langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle = \|E_\perp\|^2 + 2\|E_K\|^2 + \|B_T\|^2 + 2\|B_K\|^2 + \|B_\perp\|^2.$$

Proof. Product decomposition splits one-forms into temporal and spatial parts. Adapted temporal-mixed condition gives $\Psi(\partial_t) = 0$, so the temporal assembled connection has only the fixed tangent block and the normal component A_0 . Its spatial off-diagonal block is $\Psi(K_\Sigma)$. Since no time derivative of A_0 occurs in the curvature density, A_0 is a multiplier in the formal fixed-product Legendre transform. Normal electric component is the 0i-component of the normal curvature of $A_0 dt + A$, hence $E_\perp = \dot{A} - D_A A_0$. Since $\Psi(\partial_t) = 0$, the mixed electric component is $D_t^{A_0} K$. Magnetic fields are the purely spatial curvature blocks. Proposition 1.10, applied separately to electric and magnetic parts, gives the energy identity; the two off-diagonal mixed blocks give the coefficient 2. $\square$

Proposition 8.3 (Variational Current and Canonical Form). *Let*

$$\mathcal{L}_{\text{curv}} = \frac{1}{2} \text{tr}(\mathcal{F}^\oplus \wedge \star \mathcal{F}^\oplus)$$

denote the normalized quadratic curvature density. For an admissible variation $a = \delta\omega^\oplus$,

$$\delta\mathcal{F}^\oplus = D^\oplus a$$

and

$$\delta\mathcal{L}_{\text{curv}} = \left\langle a, (D^\oplus)^\dagger \mathcal{F}^\oplus \right\rangle dV_h + d\Theta_{\text{curv}}(a), \quad \Theta_{\text{curv}}(a) = \text{tr}(a \wedge \star \mathcal{F}^\oplus).$$

Since a ranges only over $T_{\omega^\oplus} \mathcal{A}_\oplus$, the corresponding Euler equation is $\Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0$. With respect to the curvature block decomposition and the admissible variation blocks $a = (\delta A, \delta K)$, the current has block form

$$\Theta_{\text{curv}}(a) = -\text{tr}_N(\delta A \wedge \star S) + 2\langle \delta K \wedge \star C \rangle_{\text{mix}}.$$

Here tr_N is the normal trace pairing and $\langle \cdot \rangle_{\text{mix}}$ is the induced wedge pairing on the mixed block. For admissible variations a_1, a_2 , define

$$\omega_{\text{curv}}(a_1, a_2) := \delta_2 \Theta_{\text{curv}}(a_1) - \delta_1 \Theta_{\text{curv}}(a_2).$$

For a slice $\Sigma_t \cong \Sigma$, define the formal hypersurface form

$$\Omega_{\Sigma_t}(a_1, a_2) := \int_{\Sigma_t} \omega_{\text{curv}}(a_1, a_2).$$

With the sign convention $\omega_{\text{curv}} = -\delta \Theta_{\text{curv}}$, the hypersurface form satisfies

$$\Omega_{\Sigma_t} = \int_{\Sigma} (\delta A \wedge \delta E_{\perp} + 2\delta K \wedge \delta E_K) dV_{\gamma}.$$

Proof. The covariant first-variation and presymplectic-current identities are Proposition B.1 and Proposition B.2. The block identity follows by inserting

$$a = \begin{pmatrix} 0 & -(\delta K)^* \\ \delta K & \delta A \end{pmatrix}, \quad \mathcal{F}^{\oplus} = \begin{pmatrix} G & -C^* \\ C & S \end{pmatrix}$$

into the trace pairing. Both off-diagonal mixed blocks contribute equally, giving the coefficient 2. Pulling $\star S$ and $\star C$ back to Σ_t removes the dt -magnetic parts and retains the spatial Hodge duals of $E_{\perp}$ and E_K . Hence the displayed hypersurface form follows. $\square$

Proposition 8.4 (Legendre Hamiltonian and Multiplier). *In the adapted fixed-product canonical regime, the spatially integrated normalized Riemannian curvature Lagrangian has the split form*

$$L_{\Sigma} = \int_{\Sigma} \frac{1}{2} (\|E_{\perp}\|^2 + 2\|E_K\|^2) dV_{\gamma} + V_{\text{mag}}, \quad V_{\text{mag}} = \frac{1}{2} \int_{\Sigma} (\|B_T\|^2 + 2\|B_K\|^2 + \|B_{\perp}\|^2) dV_{\gamma}.$$

Canonical momenta are

$$p_A = E_{\perp}, \quad p_K = 2E_K,$$

so the Legendre inversion is

$$E_{\perp} = p_A, \quad E_K = \frac{1}{2} p_K.$$

After separating the A_0 -multiplier term, the formal Riemannian Legendre Hamiltonian density and Hamiltonian are

$$\mathcal{H} = \frac{1}{2} \|p_A\|^2 + \frac{1}{4} \|p_K\|^2 - \frac{1}{2} (\|B_T\|^2 + 2\|B_K\|^2 + \|B_{\perp}\|^2), \quad H = \int_{\Sigma} \mathcal{H} dV_{\gamma}.$$

Formal canonical one-form and symplectic form are

$$\Theta_{\text{can}} = \int_{\Sigma} (\langle p_A, \delta A \rangle + \langle p_K, \delta K \rangle) dV_{\gamma},$$

$$\Omega_{\text{can}} = -\delta \Theta_{\text{can}} = \int_{\Sigma} (\delta A \wedge \delta p_A + \delta K \wedge \delta p_K) dV_{\gamma}.$$

Let $H_R := H$. With the internal momentum map J of Proposition 8.5, the multiplier Hamiltonian is

$$H_{\text{ext}}[A_0] = H_R + \langle J, A_0 \rangle,$$

and its Hamilton equations give

$$\dot{A} = p_A + D_A A_0, \quad \dot{K} = \frac{1}{2} p_K - A_0 \cdot K.$$

The reduced momentum equations have the Riemannian sign

$$\dot{p}_A = \frac{\delta V_{\text{mag}}}{\delta A}, \quad \dot{p}_K = \frac{\delta V_{\text{mag}}}{\delta K},$$

before adding the A_0 -generated internal gauge terms. The displayed Hamiltonian is therefore a formal Riemannian Legendre Hamiltonian, not an energy-positivity statement.

Proof. By Proposition 8.2, the velocity-dependent part of L_Σ is

$$\frac{1}{2} \int_\Sigma (\|\dot{A} - D_A A_0\|^2 + 2\|D_t^{A_0} K\|^2) dV_\gamma.$$

The velocity Hessian on $(\dot{A}, \dot{K})$ is diagonal with weights 1 and 2, giving $p_A = E_\perp$, $p_K = 2E_K$, and the stated inversion. Formal Riemannian Legendre transform gives $H_R = T^*(p) - V_{\text{mag}}$. Hence $\dot{p} = -\delta H_R / \delta q$ gives $+\delta V_{\text{mag}} / \delta q$. The term $\langle J, A_0 \rangle$ contributes the infinitesimal internal gauge action, giving the displayed A_0 -dependent velocities. Proposition 8.3 gives the canonical symplectic form after substituting $p_A = E_\perp$ and $p_K = 2E_K$. $\square$

Let

$$\mathfrak{g}_N = \Gamma(\mathfrak{so}(N)).$$

Define the Gauss expression

$$\mathcal{G}(A, K, p_A, p_K) := D_A p_A + p_K K^* - K p_K^*.$$

For $\epsilon \in \mathfrak{g}_N$, the infinitesimal internal gauge action on the formal canonical variables is

$$\delta_\epsilon A = D_A \epsilon, \quad \delta_\epsilon K = \epsilon \cdot K, \quad \delta_\epsilon p_A = [\epsilon, p_A], \quad \delta_\epsilon p_K = \epsilon \cdot p_K.$$

Proposition 8.5 (Internal Gauge Momentum Map and Gauss Constraint). *Internal gauge action on the formal canonical phase space is Hamiltonian. With respect to the algebraic dual pairing between $\mathfrak{g}_N^\vee$ and $\mathfrak{g}_N$, define*

$$\langle J(A, K, p_A, p_K), \epsilon \rangle = - \int_\Sigma \langle \epsilon, \mathcal{G}(A, K, p_A, p_K) \rangle dV_\gamma.$$

Then

$$d\langle J, \epsilon \rangle = \iota_{X_\epsilon} \Omega_{\text{can}},$$

where X_ϵ is the infinitesimal gauge vector field. Variation of the split Lagrangian with respect to A_0 gives

$$\mathcal{G}(A, K, p_A, p_K) = D_A p_A + p_K K^* - K p_K^* = 0.$$

Thus A_0 is the multiplier for the internal Gauss constraint

$$J = 0, \quad \mathcal{S}_G = J^{-1}(0).$$

Proof. Contract Ω_{can} with

$$X_\epsilon = (D_A \epsilon, \epsilon \cdot K, [\epsilon, p_A], \epsilon \cdot p_K).$$

Integration by parts in the $D_A \epsilon$ term and regrouping the normal-bundle action terms gives the differential of the displayed functional with the stated minus sign in the definition of J . Explicitly, the p_A -variation of $-\int_\Sigma \langle \epsilon, D_A p_A \rangle dV_\gamma$ is $\int_\Sigma \langle D_A \epsilon, \delta p_A \rangle dV_\gamma$, matching the A -component of $\iota_{X_\epsilon} \Omega_{\text{can}}$. Remaining terms are the cotangent-lift terms for the normal action on K , as in the standard momentum-map computation for Hamiltonian gauge actions [12, 13].

Let $\eta = \delta A_0$. Proposition 8.2 gives

$$\delta_\eta E_\perp = -D_A \eta, \quad \delta_\eta E_K = \eta \cdot K.$$

Using $p_A = E_\perp$ and $p_K = 2E_K$, the A_0 -variation of the velocity part of L_Σ is

$$\delta_\eta L_\Sigma = \int_\Sigma (\langle p_A, -D_A \eta \rangle + \langle p_K, \eta \cdot K \rangle) dV_\gamma.$$

Integrating the first term by parts and rewriting the second by the normal adjoint pairing gives

$$\delta_\eta L_\Sigma = \int_\Sigma \langle \eta, D_A p_A + p_K K^* - K p_K^* \rangle dV_\gamma.$$

Arbitrariness of η under the boundary regime gives the displayed multiplier equation. Algebraic dual pairing with arbitrary $\epsilon \in \mathfrak{g}_N$ separates sections of the normal adjoint bundle; since J is the negative dual of $\mathcal{G}$, $J = 0$ is equivalent to $\mathcal{G} = 0$. $\square$

Write

$$\star \mathcal{F}^\oplus = \mathcal{E}_\star^\oplus + dt \wedge \mathcal{B}_\star^\oplus,$$

where $\mathcal{E}_\star^\oplus$ is the spatial Hodge dual of the electric channel and $\mathcal{B}_\star^\oplus$ is the spatial Hodge dual of the magnetic channel. In the fixed product split,

$$D^\oplus \star \mathcal{F}^\oplus = dt \wedge (\partial_t \mathcal{E}_\star^\oplus + [\omega_0^\oplus, \mathcal{E}_\star^\oplus] - D_{\omega_\Sigma^\oplus} \mathcal{B}_\star^\oplus) + D_{\omega_\Sigma^\oplus} \mathcal{E}_\star^\oplus.$$

Ambient Yang–Mills expression has temporal channel

$$D_{\omega_\Sigma^\oplus} \mathcal{E}_\star^\oplus = 0$$

and spatial evolution channel

$$\partial_t \mathcal{E}_\star^\oplus + [\omega_0^\oplus, \mathcal{E}_\star^\oplus] - D_{\omega_\Sigma^\oplus} \mathcal{B}_\star^\oplus = 0.$$

Projection by $\Pi_{A_\oplus}$ retains only the normal and admissible symmetric mixed parts of these channels.

Proposition 8.6 (Projected Covariant-to-Hamiltonian Identification). *Under Assumption 8.1, and restricted to its adapted temporal-mixed sector, the normal projection of the temporal channel of $\Pi_{A_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0$ is the Gauss constraint of Proposition 8.5 after the Legendre substitution*

$$p_A = E_\perp, \quad p_K = 2E_K.$$

The spatial projected Euler equations obtained from the same adapted split are related to the Hamiltonian system by the nondegenerate Legendre transform of Proposition 8.4. Thus the fixed-product Euler system in this adapted sector has the formal Hamiltonian representation generated by $H_{\text{ext}}[A_0]$. In the gauge $A_0 = 0$, the generator is H_R . No consistency or equivalence with the unrestricted projected variational problem is asserted.

Proof. Normal block of $D_{\omega_\Sigma^\oplus} \mathcal{E}_*^\oplus = 0$ is

$$\mathcal{G}(A, K, p_A, p_K) = D_A p_A + p_K K^* - K p_K^* = 0$$

after substituting $p_A = E_\perp$ and $p_K = 2E_K$, hence agrees with the multiplier equation of Proposition 8.5. The spatial projected channel gives the A - and K -Euler equations of the fixed-product Lagrangian L_Σ . Proposition 8.4 gives a nondegenerate Legendre transform in $(E_\perp, E_K)$, and Proposition 8.4 records the resulting Riemannian sign and A_0 -multiplier terms. The standard Legendre equivalence between Euler–Lagrange and Hamilton equations then gives the displayed Hamiltonian system in the adapted sector. $\square$

Functional derivatives are taken with respect to the $L^2(dV_\gamma)$ pairing on the displayed canonical variables. Formal Poisson bracket induced by Ω_{can} is

$$\{F, G\} = \int_\Sigma \left(\left\langle \frac{\delta F}{\delta A}, \frac{\delta G}{\delta p_A} \right\rangle - \left\langle \frac{\delta F}{\delta p_A}, \frac{\delta G}{\delta A} \right\rangle + \left\langle \frac{\delta F}{\delta K}, \frac{\delta G}{\delta p_K} \right\rangle - \left\langle \frac{\delta F}{\delta p_K}, \frac{\delta G}{\delta K} \right\rangle \right) dV_\gamma.$$

Theorem 8.7 (Formal Gauge-Canonical Interface). *In the higher-rank regime $n \geq 2$, $k \geq 2$, under Assumption 8.1 and restricted to its adapted temporal-mixed sector, the admissible Yang–Mills-type equation*

$$\Pi_{\mathcal{A}^\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0$$

admits the formal fixed-product canonical representation

$$(A, K, p_A, p_K, \Omega_{\text{can}}, H_R, H_{\text{ext}}, J).$$

Internal Gauss generators close, and the reduced formal Riemannian Hamiltonian is internally gauge invariant:

$$\{\langle J, \epsilon_1 \rangle, \langle J, \epsilon_2 \rangle\} = \langle J, [\epsilon_1, \epsilon_2] \rangle, \quad \{H_R, \langle J, \epsilon \rangle\} = 0.$$

Consequently the Gauss surface $\mathcal{S}_G$ is preserved by the H_R flow and by the multiplier flow of $H_{\text{ext}}[A_0]$. Only the adapted-sector Legendre interface is asserted; no consistency or equivalence with the unrestricted projected equation is asserted.

Proof. Proposition 8.2 fixes the adapted phase variables and electric-magnetic blocks. Proposition 8.4 gives the Legendre transform, symplectic form, reduced formal Riemannian Hamiltonian, and multiplier Hamiltonian. Proposition 8.5 gives the momentum map, Gauss surface, and multiplier origin of the constraint.

Since $\langle J, \epsilon \rangle$ generates the infinitesimal internal gauge action, the Poisson bracket of two such generators is the generator of the commutator action. Fields $E_\perp, E_K, B_T, B_K, B_\perp$ transform tensorially under the internal normal-gauge action, while B_T is fixed by the tangent block; the trace and metric pairings are invariant. Hence H_R is invariant under the internal gauge action. Therefore the displayed closure identities hold. Along H_R -Hamiltonian flow,

$$\frac{d}{dt} \langle J, \epsilon \rangle = \{\langle J, \epsilon \rangle, H_R\} = 0,$$

so initial data satisfying $J = 0$ remain on $J^{-1}(0)$. For $H_{\text{ext}}[A_0] = H_R + \langle J, A_0 \rangle$, the extra term is a linear combination of Gauss generators. The closure relation makes its flow tangent to $J^{-1}(0)$. $\square$

Remark 8.8 (Formal Reduced Observable Algebra). *Internally gauge-invariant formal observables are closed under the canonical Poisson bracket by the Jacobi identity and the Gauss-generator closure relation. Their restrictions to $\mathcal{S}_G$ therefore inherit a formal reduced observable algebra. No smooth quotient or global symplectic reduction theorem is asserted.*

Remark 8.9 (Scope of the Canonical Interface). *Canonical interface here means the adapted fixed-product interface of Assumption 8.1. It does not include independent temporal mixed variables K_{00} or K_{0i} , does not prove consistency of the adapted sector under the unrestricted projected covariant equation, does not impose a Lorentzian Cauchy interpretation, and does not identify the construction with ADM canonical gravity.*

9 Codimension-One Einstein Readout

Higher normal rank leaves a normal curvature channel. Rank one removes it. The same assembled curvature then has only Gauss and mixed residual channels, and these channels admit a vector-scalar readout.

The readout is intrinsic on M . The Cauchy interpretation enters only after specializing M to a three-dimensional hypersurface Σ .

9.1 Rank-One Normal-Channel Collapse

Theorem 9.1 (Rank-One Normal-Channel Collapse). *Let $N = L$ be an oriented metric line bundle. Then $\mathfrak{so}(L) = 0$, the local normal curvature channel is absent, and*

$$S \equiv 0.$$

Consequently,

$$\mathcal{F}^\oplus = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & 0 \end{pmatrix}.$$

Mixed channel $\mathcal{C}$ and Gauss channel $\mathcal{G}$ are the surviving codimension-one curvature channels.

Proof. For an oriented metric line bundle, the oriented metric structure group is trivial and the structure Lie algebra is zero. Normal curvature $F^\perp$ and the normal block $K \wedge K^*$ are $\mathfrak{so}(L)$ -valued, hence both vanish as local curvature channels. Proposition 1.8 gives the block identity. $\square$

9.2 Residual Vector-Scalar Readout

Definition 9.2 (Codimension-One Compatibility Readout). *Let $\mathcal{A}_\oplus^{(1)} \subset \mathcal{A}_\oplus$ be the rank-one slice. For $\nabla^\oplus \in \mathcal{A}_\oplus^{(1)}$, define*

$$\text{Contr}(\mathcal{C})(X) := \sum_j \mathcal{C}(e_j, X)(e_j) \in L,$$

where (e_j) is any local h -orthonormal frame, and define

$$\text{tr}_h(\mathcal{G}) := \sum_{i,j} h(\mathcal{G}(e_i, e_j)e_j, e_i).$$

Codimension-one compatibility readout is

$$\mathcal{K}_{\text{comp}} : \mathcal{A}_\oplus^{(1)} \longrightarrow \Gamma(T^*M \otimes L) \oplus C^\infty(M), \quad \mathcal{K}_{\text{comp}}(\nabla^\oplus) = (\text{Contr}(\mathcal{C}), \text{tr}_h(\mathcal{G})).$$

Proposition 9.3 (Intrinsic Readout). *Readout $\mathcal{K}_{\text{comp}}$ is defined by the rank-one curvature blocks, the metric h , and the line metric on L . It is independent of the choice of h -orthonormal frame and is natural under oriented orthogonal line-bundle isometries.*

Proof. Both displayed contractions are complete metric contractions of tensorial curvature blocks. Orthogonal changes of the tangent frame preserve the contracted tensors. In the mixed contraction, the line factor remains as the L -valued component of $\text{Contr}(\mathbf{C})$, so no choice of unit normal is required for the intrinsic definition. $\square$

Definition 9.4 (Fixed-Metric Vacuum Constraint Expressions). *After choosing a unit section of L , the rank-one mixed tensor is represented by a symmetric two-tensor K . Define*

$$\mathcal{K}_{\text{vac}}^{(h)}(K) := (\nabla^j K_{ij} - \nabla_i \text{tr}_h K, \text{scal}_h - \|K\|_h^2 + (\text{tr}_h K)^2).$$

Here h is held fixed in the readout, while the physical Cauchy interpretation later regards (Σ, h, K) as initial data.

Theorem 9.5 (Fixed-Metric Vacuum Constraint Identification). *Assume $N = L$ has rank one and choose a unit-normal trivialization. Under this trivialization, $\mathcal{K}_{\text{comp}}$ is represented by*

$$\mathcal{K}_{\text{vac}}^{(h)}.$$

Thus the codimension-one compatibility readout is the standard fixed-metric vacuum Einstein constraint pair.

Proof. In a unit-normal trivialization, K is an ordinary symmetric two-tensor. Mixed curvature has components

$$\mathbf{C}_{ijk} = \nabla_i K_{jk} - \nabla_j K_{ik},$$

with antisymmetry in the first two slots and the cyclic identity inherited from the symmetry of K . Contracting the first and third slots in the definition of $\text{Contr}(\mathbf{C})$ gives

$$(\text{Contr}(\mathbf{C}))_i = h^{jk} \mathbf{C}_{jik} = \nabla^j K_{ij} - \nabla_i \text{tr}_h K.$$

For the Gauss block,

$$\mathbf{G} = R^h - K^* \wedge K.$$

With the wedge convention of Proposition 1.8, scalar contraction gives

$$\text{tr}_h(\mathbf{G}) = \text{scal}_h - \|K\|_h^2 + (\text{tr}_h K)^2.$$

These are the standard fixed-metric vacuum constraint expressions in the Cauchy formulation of the Einstein equations [5, 6]. $\square$

Remark 9.6 (Relation to the Standard Constraint Derivation). *Classical hypersurface derivations obtain the same vector-scalar pair by projecting the Einstein tensor along a spacetime normal [5, 6]. Theorem 9.5 obtains it as the rank-one residual contraction of $\mathcal{F}^\oplus$. The standard Cauchy interpretation enters after this curvature-readout identification.*

9.3 Natural Linear Metric Classification

Remark 9.7 (Linearity of the Rank-One Readout). *In rank one, a unit-normal trivialization identifies the normal line with the scalar normal direction used in the Cauchy formulation. The standard vacuum momentum and Hamiltonian constraints are linear metric contractions of the residual mixed and Gauss curvature channels. Definition 9.8 isolates this readout category. Theorem 9.9 classifies its natural vector-scalar outputs, and Theorem 9.5 identifies the unique nondegenerate output with the fixed-metric vacuum Einstein constraint pair.*

Definition 9.8 (Rank-One Linear Contraction Category). *Let $V = T_x M$ be an n -dimensional inner-product space with $n \geq 2$, and let L be an oriented metric line. Let*

$$\mathcal{W}_L \subset \Lambda^2 V^* \otimes V^* \otimes L$$

denote the mixed Codazzi-type fiber consisting of tensors C_{ijk} satisfying

$$C_{ijk} = -C_{jik}, \quad C_{ijk} + C_{jki} + C_{kij} = 0,$$

and let $\mathcal{R} \subset \Lambda^2 V^ \otimes \mathfrak{so}(V)$ be the algebraic curvature-tensor fiber. A natural linear metric vector-scalar output is an $O(V)$ -equivariant linear map*

$$\mathcal{W}_L \oplus \mathcal{R} \longrightarrow (V^* \otimes L) \oplus \mathbb{R}$$

formed using only the metric contractions in V and the metric line L . Orientation tensors, volume forms, and dimension-specific ε -contractions are not objects of this category.

Theorem 9.9 (Rank-One Natural Contraction Classification). *For $n \geq 2$, in the linear, metric, $O(V)$ -equivariant category of Definition 9.8, contractions $\mathcal{W}_L \rightarrow V^* \otimes L$ are scalar multiples of*

$$C \longmapsto h^{jk} C_{jik};$$

contractions $\mathcal{R} \rightarrow \mathbb{R}$ are scalar multiples of scalar-curvature contraction; and no metric contractions $\mathcal{W}_L \rightarrow \mathbb{R}$ or $\mathcal{R} \rightarrow V^ \otimes L$ exist. Orientation tensors, nonlinear operations, and higher-order operators are outside the category.*

Proof. Appendix A proves the linear metric contraction classification. Standard natural-operator and orthogonal-invariant contraction principles provide the invariant-theoretic background [8, 18]. Oriented ε -contractions and other dimension-specific operations are outside Definition 9.8. $\square$

Theorem 9.10 (Codimension-One Readout Determination). *In the rank-one natural linear metric vector-scalar category, every output has the form*

$$\Theta(C, G) = (a \operatorname{Contr}(C), b \operatorname{tr}_h(G))$$

for constants a, b . In the nondegenerate case $a \neq 0$ and $b \neq 0$, its zero locus agrees with the zero locus of $\mathcal{K}_{\text{comp}}$. After unit-normal trivialization this is the fixed-metric vacuum Einstein constraint locus

$$\mathcal{K}_{\text{vac}}^{(h)}(K) = 0.$$

Proof. Theorem 9.1 leaves only the mixed and Gauss channels. Theorem 9.9 gives one vector output from the mixed channel, one scalar output from the Gauss channel, and no cross-channel outputs. Hence every vector-scalar output has the displayed form. If a and b are both nonzero, its vanishing is equivalent to the simultaneous vanishing of $\operatorname{Contr}(C)$ and $\operatorname{tr}_h(G)$. Theorem 9.5 gives the final identification. $\square$

Remark 9.11 (Degenerate Outputs). *If $a = 0$ or $b = 0$, the output imposes only the corresponding scalar or vector component. Nondegenerate vector-scalar readout therefore gives the full fixed-metric vacuum constraint locus.*

Definition 9.12 (Physical Rank-One Cauchy Regime). *Physical rank-one Cauchy regime means that $M = \Sigma$ is a smooth three-dimensional manifold without boundary, h is a smooth Riemannian metric on Σ , $N = L$ is an oriented metric line bundle, and a unit-normal trivialization identifies the mixed tensor with a smooth symmetric two-tensor K on Σ .*

Proposition 9.13 (Initial-Data Interpretation). *In the physical rank-one Cauchy regime, if*

$$\mathcal{K}_{\text{comp}}(\nabla^\oplus) = 0,$$

then (Σ, h, K) is a smooth vacuum Einstein initial data set in the standard Cauchy-problem sense.

Proof. Theorem 9.5 represents $\mathcal{K}_{\text{comp}}$ by $\mathcal{K}_{\text{vac}}^{(h)}$ after unit-normal trivialization. Its vanishing is precisely the vacuum momentum and Hamiltonian constraint system for a smooth initial data set (Σ, h, K) . $\square$

Corollary 9.14 (Cauchy Handoff). *In the physical rank-one Cauchy regime, assume $\mathcal{K}_{\text{comp}}(\nabla^\oplus) = 0$. Then, by the standard Choquet-Bruhat–Geroch Cauchy theorem, there is a maximal globally hyperbolic vacuum development of the initial data (Σ, h, K) . This development is unique among globally hyperbolic vacuum developments of the same initial data, up to an isometry preserving the embedded Cauchy hypersurface and the induced initial data.*

Proof. Proposition 9.13 supplies the vacuum constraint-satisfying initial data set. Standard Einstein Cauchy theory gives existence and uniqueness of the maximal globally hyperbolic vacuum development [7, 6]. The compatibility-curvature identification with the vacuum constraint map is proved above; the Lorentzian development theorem is imported from standard Cauchy theory. $\square$

10 Conclusion

No separate curvature object is assigned to either sector. A single metric connection on $TM \oplus N$ supplies one curvature whose rank-one collapse gives the vacuum constraint readout and whose higher-rank normal channel supports the canonical gauge sector.

Classification is the decisive step. Field content is not introduced by independently choosing an Einstein sector and a Yang–Mills sector. It is read from split metric curvature after admissible contraction, first-jet exhaustion, trace rigidity, and Legendre splitting.

What remains is a classical canonical field structure attached to a split metric bundle. In rank one it reaches the vacuum Cauchy data from which spacetime develops; in higher normal rank it reaches the internal gauge phase space with Hamiltonian and Gauss momentum map, placing both outputs in canonical form.

A Rank-One Linear Metric Contractions

This appendix verifies the linear metric $O(V)$ -equivariant contraction statements used in Theorem 9.9. Let V be an n -dimensional inner-product space with $n \geq 2$, let L be an oriented metric line, and let

$$\mathcal{W}_L \subset \Lambda^2 V^* \otimes V^* \otimes L$$

be the mixed Codazzi-type fiber of Definition 9.8. Weyl invariant theory reduces this category to contractions formed from the metric tensor and its inverse [18]. Orientation tensors, nonlinear operations, and higher-order operators are not part of the category audited here.

A.1 Mixed Channel to Vector

After choosing a unit vector in L , write the scalar representative of $C \in \mathcal{W}_L$ as C_{ijk} . A linear metric contraction to a covector must contract two of the three tangent covariant indices and leave

one index free. Up to relabeling, the candidates are

$$A_i = h^{jk} C_{jik}, \quad B_i = h^{jk} C_{ijk}, \quad D_i = h^{jk} C_{jki}.$$

Antisymmetry in the first two slots gives $B_i = -A_i$. Cyclic identity gives

$$C_{jki} + C_{kij} + C_{ijk} = 0.$$

Contracting by h^{jk} gives

$$D_i + h^{jk} C_{kij} + B_i = 0.$$

Since $h^{jk} C_{kij} = h^{kj} C_{jik} = A_i$, and $B_i = -A_i$, it follows that $D_i = 0$. Thus all metric vector contractions are scalar multiples of A_i . Restoring the line factor gives the generator

$$C \mapsto h^{jk} C_{jik} \in V^* \otimes L.$$

Consequently,

$$\dim \text{Hom}_{O(V)}(\mathcal{W}_L, V^* \otimes L) = 1$$

inside the linear metric contraction category.

A.2 Gauss Channel to Scalar

Let R_{ijkl} be an algebraic curvature tensor. A linear metric scalar contraction must pair all four tangent indices. Up to curvature symmetries, the possible full contractions are

$$h^{ik} h^{jl} R_{ijkl}, \quad h^{ij} h^{kl} R_{ijkl}, \quad h^{il} h^{jk} R_{ijkl}.$$

Middle expression vanishes by antisymmetry in i, j . Last expression agrees with the first up to sign by antisymmetry in the last pair and the standard curvature-pair symmetry. Hence the scalar-curvature contraction spans $\text{Hom}_{O(V)}(\mathcal{R}, \mathbb{R})$. Consequently,

$$\dim \text{Hom}_{O(V)}(\mathcal{R}, \mathbb{R}) = 1$$

inside the linear metric contraction category.

A.3 Vanishing of Mixed-to-Scalar Contractions

A scalar output from $C \in \mathcal{W}_L$ would require contracting all three tangent covariant indices using only the metric on V . Metric contractions remove tangent indices in pairs, so one tangent index remains. Therefore no $O(V)$ -equivariant linear metric contraction $\mathcal{W}_L \rightarrow \mathbb{R}$ exists. Intrinsically, the remaining L -factor also has no canonical removal in this category. Thus

$$\text{Hom}_{O(V)}(\mathcal{W}_L, \mathbb{R}) = 0.$$

A.4 Vanishing of Gauss-to-Vector Contractions

A vector output from one algebraic curvature tensor would require leaving exactly one tangent covariant index unpaired after metric contraction. Since the curvature tensor has four covariant indices and metric contractions remove indices in pairs, no such linear metric contraction exists. Gauss channel also contains no L -factor, so it cannot produce a $V^* \otimes L$ output in the intrinsic rank-one category. Thus

$$\text{Hom}_{O(V)}(\mathcal{R}, V^* \otimes L) = 0.$$

The four cases exhaust all vector-scalar outputs in Definition 9.8.

B Covariant Variation and Internal Gauge Identity

This appendix records the covariant first-variation, presymplectic-current, and restricted internal-gauge identities used in Sections 5 and 8. The current convention is the standard covariant phase-space convention [9, 10]. Section 8 applies these identities after a fixed-product split; the identities below are the covariant source.

Proposition B.1 (Yang–Mills First Variation and Potential Current). *For the classified Yang–Mills density*

$$L_{\text{YM}} = \frac{1}{2} \text{tr}(\mathcal{F}^\oplus \wedge \star \mathcal{F}^\oplus) = \frac{1}{2} \langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle dV_h,$$

the first variation is

$$\delta L_{\text{YM}} = \left\langle \delta\omega^\oplus, (D^\oplus)^\dagger \mathcal{F}^\oplus, \right\rangle dV_h + d\Theta_L(\delta\omega^\oplus),$$

with

$$\Theta_L(\delta\omega^\oplus) = \text{tr}(\delta\omega^\oplus \wedge \star \mathcal{F}^\oplus).$$

The sign convention is fixed by

$$\mathcal{E}_{\text{amb}} = -\star D^\oplus(\star \mathcal{F}^\oplus).$$

Under this adjoint convention, the ambient equation $(D^\oplus)^\dagger \mathcal{F}^\oplus = 0$ is equivalent to $D^\oplus \star \mathcal{F}^\oplus = 0$. On the restricted configuration space $\mathcal{A}_\oplus$, stationarity is instead the projected equation $\Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0$.

Proof. Using $\delta \mathcal{F}^\oplus = D^\oplus(\delta\omega^\oplus)$,

$$\delta L_{\text{YM}} = \text{tr}(\delta \mathcal{F}^\oplus \wedge \star \mathcal{F}^\oplus) = \text{tr}(D^\oplus(\delta\omega^\oplus) \wedge \star \mathcal{F}^\oplus).$$

Covariant integration by parts in the boundary regime fixed in the main text gives

$$\text{tr}(D^\oplus(\delta\omega^\oplus) \wedge \star \mathcal{F}^\oplus) = \left\langle \delta\omega^\oplus, (D^\oplus)^\dagger \mathcal{F}^\oplus, \right\rangle dV_h + d \text{tr}(\delta\omega^\oplus \wedge \star \mathcal{F}^\oplus).$$

Hence

$$\delta L_{\text{YM}} = \left\langle \delta\omega^\oplus, (D^\oplus)^\dagger \mathcal{F}^\oplus, \right\rangle dV_h + d\Theta_L(\delta\omega^\oplus),$$

with $\Theta_L(\delta\omega^\oplus) = \text{tr}(\delta\omega^\oplus \wedge \star \mathcal{F}^\oplus)$. □

Proposition B.2 (Covariant Presymplectic Current for the Yang–Mills Channel). *Covariant presymplectic current is*

$$\omega_L(\delta_1\omega^\oplus, \delta_2\omega^\oplus) := \delta_2\Theta_L(\delta_1\omega^\oplus) - \delta_1\Theta_L(\delta_2\omega^\oplus),$$

and for the classified Yang–Mills channel this specializes to

$$\omega_L(\delta_1, \delta_2) = \delta_2 \text{tr}(\delta_1\omega^\oplus \wedge \star \mathcal{F}^\oplus) - \delta_1 \text{tr}(\delta_2\omega^\oplus \wedge \star \mathcal{F}^\oplus).$$

Proof. This is the canonical sign convention $\omega_L = -\delta\Theta_L$ used in the main text. Substituting $\Theta_L(\delta\omega^\oplus) = \text{tr}(\delta\omega^\oplus \wedge \star \mathcal{F}^\oplus)$ yields the displayed Yang–Mills specialization. □

Theorem B.3 (Internal Gauge Noether Identity and Yang–Mills Specialization). *Let $\pi_{\mathfrak{so}(N)}$ denote normal-block projection in the splitting $E = TM \oplus N$. For any smooth local Lagrangian on the restricted configuration space $\mathcal{A}_\oplus$ invariant under the restricted internal normal gauge group, let $\mathcal{E}_L$ denote its restricted Euler–Lagrange operator, embedded in $\Omega^1(M, \mathfrak{so}(E))$ through $T_{\omega^\oplus} \mathcal{A}_\oplus \subset \Omega^1(M, \mathfrak{so}(E))$. Then*

$$\pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_L) = 0.$$

For the classified Yang–Mills channel:

1. the Euler–Lagrange equation on the restricted configuration space is

$$\Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0;$$

2. with projected Euler operator $\mathcal{E}_{\text{YM}} = \Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus)$, restricted gauge symmetry $\delta_\xi \omega^\oplus = D^\oplus \xi$ yields the specialized projected Noether identity

$$\pi_{\mathfrak{so}(N)}((D^\oplus)^\dagger \mathcal{E}_{\text{YM}}) = 0.$$

Proof. Item (1) follows from Proposition B.1 and arbitrariness of admissible variations in $T_{\omega^\oplus} \mathcal{A}_\oplus$. For the symmetry identity, the restricted normal gauge action has infinitesimal generator $D^\oplus \xi \in T_{\omega^\oplus} \mathcal{A}_\oplus$ for each $\xi \in \Gamma(\mathfrak{so}(N))$. Pairing the restricted Euler operator with this admissible generator uses its canonical embedding in the ambient $\mathfrak{so}(E)$ -valued one-form bundle before $(D^\oplus)^\dagger$ is applied. Gauge invariance gives

$$0 = \delta_\xi S = \int_M \langle \mathcal{E}_L, D^\oplus \xi \rangle dV_h.$$

Covariant integration by parts yields

$$0 = - \int_M \langle (D^\oplus)^\dagger \mathcal{E}_L, \xi \rangle dV_h,$$

and arbitrariness of the normal gauge parameter $\xi \in \Gamma(\mathfrak{so}(N))$ implies the normal-block projected identity. Applying this to $\mathcal{E}_L = \mathcal{E}_{\text{YM}}$ gives item (2). $\square$

C Projected Euler Block Form

This appendix records the block form of the projected Euler equation

$$\Pi_{\mathcal{A}_\oplus}((D^\oplus)^\dagger \mathcal{F}^\oplus) = 0$$

used in Section 4. The Euler operator and first variation are derived in Section 4 and Appendix B; only the $TM \oplus N$ block expansion is recorded here.

Proposition C.1 (Admissible Projected Euler Block Form). *Using the curvature decomposition of Proposition 1.8, define*

$$Y_T := \star(R^h - K^* \wedge K), \quad Y_M := \star C, \quad Y_N := \star(F^\perp - K \wedge K^*).$$

Then

$$\star \mathcal{F}^\oplus = \begin{pmatrix} Y_T & -Y_M^* \\ Y_M & Y_N \end{pmatrix}.$$

Writing

$$D^\oplus = D^{\text{prod}} + [\Psi, \cdot], \quad \Psi = \begin{pmatrix} 0 & -K^* \\ K & 0 \end{pmatrix},$$

the projected Euler equation is the normal block equation together with the symmetric admissible mixed projection:

$$\begin{aligned} 0 &= (D^\oplus \star \mathcal{F}^\oplus)_{NN} = D^\perp Y_N + \pi_N([\Psi, \star \mathcal{F}^\oplus]), \\ 0 &= \Pi_{\text{sym}}((D^\oplus \star \mathcal{F}^\oplus)_{\text{mix}}) = \Pi_{\text{sym}}(D^{h,\perp} Y_M + \pi_{\text{mix}}([\Psi, \star \mathcal{F}^\oplus])). \end{aligned}$$

Here π_N and π_{mix} are the normal and off-diagonal mixed block projections. Frozen tangent block and skew mixed block are not Euler equations of the restricted variational problem.

Proof. The Hodge star acts on form degree and preserves endomorphism block type, giving the displayed block form of $\star\mathcal{F}^\oplus$. Product derivative D^{prod} acts diagonally on the splitting, producing $D^h Y_T$, $D^{h,\perp} Y_M$, and $D^\perp Y_N$. The remaining contribution is the graded commutator with the off-diagonal field Ψ . Admissible projection keeps the normal block and the symmetric mixed part, while discarding the frozen tangent and skew mixed directions by Definition 1.6. $\square$

D Mixed Raw Symbol and Restricted Gauge Image

This appendix supplies the mixed-kernel and restricted-gauge symbol calculations used in Proposition 6.1. Fix a block-diagonal compatible background

$$K_0 = 0, \quad F_0^\perp = 0.$$

Variations may first be computed in the auxiliary unconstrained mixed B -model

$$V_B \simeq \Omega^1(M, \mathfrak{so}(N)) \oplus \Omega^1(M, \text{Hom}(TM, N)),$$

and then projected to the admissible subbundle

$$V_{\mathcal{A}^\oplus} \simeq \Omega^1(M, \mathfrak{so}(N)) \oplus \Gamma(\text{Sym}^2 T^* M \otimes N).$$

At principal-symbol level,

$$\sigma_\xi(D_0^\oplus)(a) = \xi \wedge a, \quad \sigma_\xi((D^\oplus)_0^\dagger)(b) = \iota_{\xi^\sharp} b,$$

so the raw exterior-symbol composition on the auxiliary bundle is

$$a \longmapsto \iota_{\xi^\sharp}(\xi \wedge a) = |\xi|_h^2 a - \xi \otimes \iota_{\xi^\sharp} a.$$

Lemma D.1 (Projected Raw Mixed Symbol). *Let $V_{\text{mix},x} = \text{Sym}^2 T_x^* M \otimes N_x$. At the block-diagonal compatible background, the raw mixed principal symbol of the projected Euler operator is*

$$\sigma_\xi^{K,\text{raw}}(a_K) = \Pi_{\text{sym}} \iota_{\xi^\sharp}(\xi \wedge a_K) = \Pi_{\text{sym}}(|\xi|_h^2 a_K - \xi \otimes \iota_{\xi^\sharp} a_K),$$

for $a_K \in V_{\text{mix},x}$. In components,

$$(\sigma_\xi^{K,\text{raw}} a_K)_{ij}^\alpha = |\xi|_h^2 (a_K)_{ij}^\alpha - \frac{1}{2} (\xi_i \xi^p (a_K)_{pj}^\alpha + \xi_j \xi^p (a_K)_{pi}^\alpha).$$

For $\xi \neq 0$,

$$\ker \sigma_\xi^{K,\text{raw}} = \{\xi \otimes \xi \otimes \nu : \nu \in N_x\} \subset \text{Sym}^2 T_x^* M \otimes N_x.$$

Proof. On the auxiliary B -model the raw exterior-symbol composition is

$$a_B \longmapsto |\xi|_h^2 a_B - \xi \otimes \iota_{\xi^\sharp} a_B.$$

The admissible Euler operator applies the symmetric projection in the two tangent slots after this auxiliary computation, giving the displayed identity. Choose an orthonormal frame with $\xi = |\xi|_h e^1$. The projected-symbol equations become

$$(\sigma_\xi^{K,\text{raw}} a_K)_{AB} = |\xi|_h^2 (a_K)_{AB}, \quad (\sigma_\xi^{K,\text{raw}} a_K)_{1A} = \frac{1}{2} |\xi|_h^2 (a_K)_{1A}, \quad (\sigma_\xi^{K,\text{raw}} a_K)_{11} = 0,$$

for $A, B > 1$. Hence the only free component is $(a_K)_{11}^\alpha$, which is exactly the subspace $\xi \otimes \xi \otimes N_x$. $\square$

Lemma D.2 (Restricted Gauge Symbol in Admissible Blocks). *For the restricted embedding $\widehat{u} = \text{diag}(I_{T_M}, u)$, the infinitesimal gauge action is*

$$\delta_\epsilon \omega^\oplus = D^\oplus \epsilon, \quad \epsilon \in \Gamma(\mathfrak{so}(N)).$$

At a block-diagonal compatible background with $K_0 = 0$ and $F_0^\perp = 0$, its principal symbol is

$$\sigma_\xi(d)(\epsilon) = (\xi \otimes \epsilon, 0) \in T_x^* M \otimes \mathfrak{so}(N_x) \oplus \text{Sym}^2 T_x^* M \otimes N_x.$$

Thus gauge-image directions lie entirely in the normal block and have no mixed principal component.

Proof. The restricted gauge group acts only in the normal block. With block-diagonal background connection, the principal part of $D^\oplus \epsilon$ is $d\epsilon$, giving $\xi \otimes \epsilon$ in the normal component. Mixed contributions contain the mixed compatibility block and therefore vanish at principal-symbol level when $K_0 = 0$. $\square$

Consequently the mixed raw kernel $\xi \otimes \xi \otimes N_x$ is not generated by internal gauge. Adding the complementary codifferential-symbol composition $\xi \wedge \iota_{\xi^\#}$ gives the auxiliary Hodge–Laplace representative with scalar principal symbol $|\xi|_h^2 \text{Id}$, as used in Proposition 6.1.

E Curvature-Momentum Module

This appendix records the module containment used in Remark 5.3. The internal Noether identity is proved separately in Appendix B; only curvature-momentum containment is treated here.

Definition E.1 (Curvature Momentum). *For a smooth local gauge-invariant curvature density $L = \ell(\mathcal{F}^\oplus) dV_h$, define its metric curvature momentum*

$$Q_L \in \Omega^2(M, \text{End}(E))$$

by

$$\delta L = \langle \delta \mathcal{F}^\oplus, Q_L \rangle dV_h.$$

Equivalently, the form momentum is $P_L = \star Q_L \in \Omega^{n-2}(M, \text{End}(E))$. For the quadratic Yang–Mills density,

$$L_{\text{YM}} = \frac{1}{2} \langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle dV_h, \quad Q_{L_{\text{YM}}} = \mathcal{F}^\oplus, \quad P_{L_{\text{YM}}} = \star \mathcal{F}^\oplus.$$

Definition E.2 (Algebraic Curvature-Momentum Module). *Define*

$$\mathcal{J}_{\text{curv}} := \left\{ \sum_i \alpha_i \wedge Q_{L_i} \wedge \beta_i : \alpha_i, \beta_i \in \Omega^\bullet(M, \text{End}(E)), L_i = L_i(\mathcal{F}^\oplus) \right\} \subset \Omega^\bullet(M, \text{End}(E)).$$

This is the graded two-sided module over $\Omega^\bullet(M, \text{End}(E))$ algebraically generated by curvature momenta under finite sums and module operations.

Definition E.3 (Differential Curvature-Momentum Module). *Define $\mathcal{I}_{\text{curv}}$ as the smallest graded two-sided module containing $\mathcal{J}_{\text{curv}}$ and stable under both $D^\oplus$ and $(D^\oplus)^\dagger$:*

$$\mathcal{I}_{\text{curv}} := \bigcap \{ \mathcal{M} \subset \Omega^\bullet(M, \text{End}(E)) : \mathcal{J}_{\text{curv}} \subset \mathcal{M}, D^\oplus \mathcal{M} \subset \mathcal{M}, (D^\oplus)^\dagger \mathcal{M} \subset \mathcal{M} \}.$$

Proposition E.4 (Ambient and Projected Euler Divergence Form for Curvature Lagrangians). *For any smooth local gauge-invariant curvature density $L = \ell(\mathcal{F}^\oplus) dV_h$ with metric curvature momentum Q_L , the ambient Euler channel is*

$$\mathcal{E}_{\text{amb},L} = (D^\oplus)^\dagger Q_L.$$

On the restricted configuration space $\mathcal{A}_\oplus$, the Euler–Lagrange operator is

$$\mathcal{E}_L = \Pi_{\mathcal{A}_\oplus} \mathcal{E}_{\text{amb},L}.$$

For the quadratic Yang–Mills density $L_{\text{YM}} = \frac{1}{2} \langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \rangle dV_h$,

$$\mathcal{E}_L = \mathcal{E}_{\text{YM}} = \Pi_{\mathcal{A}_\oplus} ((D^\oplus)^\dagger \mathcal{F}^\oplus) = \Pi_{\mathcal{A}_\oplus} (-\star D^\oplus (\star \mathcal{F}^\oplus)).$$

Proof. For a curvature-dependent Lagrangian the connection enters the Euler channel through the curvature variation. Using $\delta \mathcal{F}^\oplus = D^\oplus(\delta \omega^\oplus)$ and the boundary regime fixed in the main text,

$$\delta \int_M L(\mathcal{F}^\oplus) = \int_M \langle D^\oplus(\delta \omega^\oplus), Q_L \rangle dV_h = \int_M \langle \delta \omega^\oplus, (D^\oplus)^\dagger Q_L \rangle dV_h.$$

Thus the ambient Euler channel is $(D^\oplus)^\dagger Q_L$. Restricting variations to $T_{\omega^\oplus} \mathcal{A}_\oplus$ gives the admissible projection. For L_{YM} , $Q_L = \mathcal{F}^\oplus$, and the fixed adjoint convention gives $(D^\oplus)^\dagger \mathcal{F}^\oplus = -\star D^\oplus (\star \mathcal{F}^\oplus)$. $\square$

Proposition E.5 (Euler Field Containment). *For any admissible curvature Lagrangian, the ambient Euler expression belongs to the curvature-momentum-generated differential module:*

$$\mathcal{E}_{\text{amb},L} \in \mathcal{I}_{\text{curv}}.$$

On $\mathcal{A}_\oplus$, the restricted Euler–Lagrange operator is its admissible projection

$$\mathcal{E}_L = \Pi_{\mathcal{A}_\oplus} \mathcal{E}_{\text{amb},L}.$$

For $L = L_{\text{YM}}$, this specializes to

$$\mathcal{E}_L = \mathcal{E}_{\text{YM}} = \Pi_{\mathcal{A}_\oplus} ((D^\oplus)^\dagger \mathcal{F}^\oplus).$$

Proof. By construction, $Q_L \in \mathcal{J}_{\text{curv}} \subseteq \mathcal{I}_{\text{curv}}$. Proposition E.4 gives

$$\mathcal{E}_{\text{amb},L} = (D^\oplus)^\dagger Q_L.$$

Definition E.3 makes $\mathcal{I}_{\text{curv}}$ stable under $(D^\oplus)^\dagger$, hence $(D^\oplus)^\dagger Q_L \in \mathcal{I}_{\text{curv}}$. $\square$

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